Online Resource 1

Quantum Optimization for Adaptive Low-Rank CP Decomposition

Full result tables

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S1 Conventions

Every table below is generated from the per-run records of the code repository (`https://github.com/vantnprof/QALR-CPD`, folder `results/revision/`) by the command `python main.py tables`. Entries are mean ± standard deviation over seeds; n is the number of runs.

- *Penalty.* μ is dimensionless; every solver receives the QUBO with $\mu_{\text{eff}} = \mu\|\mathcal{X}\|_F^2/R_{\max}$. Penalty grid of the selection protocol (Sec. 3.7): 0.005, 0.01, 0.02, 0.05, 0.1, 0.15, 0.2, 0.3, 0.4, 0.5, 0.65, 0.8, 1, 1.5, 2. A plateau is selected only if it spans at least three consecutive grid values; μ is then its middle value (the upper of the two middle values for an even number).
- *Datasets.* `synth_default`: default synthetic benchmark (20^4, $R_{\text{true}} = 6$, $\rho = 0.5$, $\sigma = 10^{-2}$); `amino`: Amino Acids; `kinetic`: X60t (error over observed entries); `covid`: COVID-19 systems serology; `snr`s`dB_rho0.5`: exact-SNR grid; `rho`r`_snr`s`dB`: SNR-matched collinearity designs; `noise_`σ and `rho_`r: noise and collinearity sweeps of the first version.
- *Methods.* As in the manuscript: ALS at fixed rank; QALR-CPD with exact search (ES), bit flipping (BF), multi-start bit flipping (MS-BF), simulated annealing (SA), tabu search (Tabu) or QAOA; in-loop magnitude pruning (Mag-IL) and its z-form variant (Diag-x); post-hoc truncation without (TopK-PH) and with 50 refit sweeps (TopK-PH+refit); in-loop truncation matched call by call to ES (TopK-IL).
- *Excess over $\varepsilon^\star$*: relative error minus the realized noise floor $\|\mathcal{N}\|_F/\|\mathcal{X}\|_F$, in percent of the floor. $\hat{R} = R_{\text{true}}$: share of seeds with exact rank recovery.

S2 Controlled ablation (manuscript Table 9)

Table S1. Default synthetic benchmark, all three penalties of the first version.

Method	Rel. error	$\hat{R}$	Excess over $\varepsilon^\star$	$\hat{R} = R_{\text{true}}$	n
synth_default, $\mu = 0.02$					
ALS ($R = 20$)	0.8129 ± 0.01167	20 ± 0	-0.7%	0%	20
TopK-PH	0.8203 ± 0.0116	7.05 ± 1.05	0.2%	40%	20
TopK-PH+refit	0.8197 ± 0.01179	7.05 ± 1.05	0.1%	40%	20
TopK-IL	0.8197 ± 0.01178	7.05 ± 1.05	0.1%	40%	20
Mag-IL	0.8197 ± 0.01176	7.15 ± 1.14	0.1%	40%	20
Diag-x	0.8188 ± 0.01184	8.9 ± 1.94	0.0%	5%	20
ES	0.8197 ± 0.01178	7.05 ± 1.05	0.1%	40%	20
BF	0.8197 ± 0.01178	7.05 ± 1.05	0.1%	40%	20
MS-BF	0.8197 ± 0.01178	7.05 ± 1.05	0.1%	40%	20
SA	0.8197 ± 0.01178	7.05 ± 1.05	0.1%	40%	20
Tabu	0.8197 ± 0.01178	7.05 ± 1.05	0.1%	40%	20
synth_default, $\mu = 0.1$					
ALS ($R = 20$)	0.8129 ± 0.01167	20 ± 0	-0.7%	0%	20
TopK-PH	0.8209 ± 0.01163	6.55 ± 0.686	0.3%	55%	20

Method	Rel. error	$\hat{R}$	Excess over $\varepsilon^\star$	$\hat{R} = R_{\text{true}}$	n
TopK-PH+refit	0.82 ± 0.01174	6.55 ± 0.686	0.2%	55%	20
TopK-IL	0.82 ± 0.01174	6.55 ± 0.686	0.2%	55%	20
Mag-IL	0.82 ± 0.01174	6.55 ± 0.686	0.2%	55%	20
Diag-x	0.8199 ± 0.01181	6.7 ± 0.923	0.2%	55%	20
ES	0.82 ± 0.01174	6.55 ± 0.686	0.2%	55%	20
BF	0.82 ± 0.01174	6.55 ± 0.686	0.2%	55%	20
MS-BF	0.82 ± 0.01174	6.55 ± 0.686	0.2%	55%	20
SA	0.82 ± 0.01174	6.55 ± 0.686	0.2%	55%	20
Tabu	0.82 ± 0.01174	6.55 ± 0.686	0.2%	55%	20
synth_default, $\mu = 0.9$					
ALS ($R = 20$)	0.8129 ± 0.01167	20 ± 0	-0.7%	0%	20
TopK-PH	0.9112 ± 0.02467	2.75 ± 0.91	11.3%	0%	20
TopK-PH+refit	0.8753 ± 0.01942	2.75 ± 0.91	6.9%	0%	20
TopK-IL	0.8755 ± 0.01973	2.75 ± 0.91	7.0%	0%	20
Mag-IL	0.9111 ± 0.0226	1 ± 0	11.3%	0%	20
Diag-x	0.8306 ± 0.01957	5.35 ± 0.745	1.5%	50%	20
ES	0.8755 ± 0.01974	2.75 ± 0.91	7.0%	0%	20
BF	0.8767 ± 0.0195	2.7 ± 0.865	7.1%	0%	20
MS-BF	0.8755 ± 0.01974	2.75 ± 0.91	7.0%	0%	20
SA	0.8755 ± 0.01974	2.75 ± 0.91	7.0%	0%	20
Tabu	0.8755 ± 0.01974	2.75 ± 0.91	7.0%	0%	20

Table S2. Real tensors, all three penalties of the first version.

Method	Rel. error	$\hat{R}$	n
amino, $\mu = 0.05$			
ALS ($R = 20$)	$0.00974 \pm 6.144\times10^{-5}$	20 ± 0	20
TopK-PH	0.04216 ± 0.0368	18.9 ± 1.35	20
TopK-PH+refit	$0.009941 \pm 3.396\times10^{-4}$	18.9 ± 1.35	20
TopK-IL	$0.009917 \pm 2.908\times10^{-4}$	18.9 ± 1.35	20
Mag-IL	$0.009853 \pm 1.479\times10^{-4}$	19.1 ± 0.912	20
Diag-x	0.02025 ± 0.002461	5.65 ± 1.27	20
ES	$0.00989 \pm 1.999\times10^{-4}$	18.9 ± 1.35	20
BF	0.06569 ± 0.1492	5.95 ± 2.61	20
MS-BF	$0.00989 \pm 1.999\times10^{-4}$	18.9 ± 1.35	20
SA	$0.00989 \pm 1.999\times10^{-4}$	18.9 ± 1.35	20
Tabu	$0.00989 \pm 1.999\times10^{-4}$	18.9 ± 1.35	20
amino, $\mu = 0.55$			
ALS ($R = 20$)	$0.00974 \pm 6.144\times10^{-5}$	20 ± 0	20
TopK-PH	0.6605 ± 0.1726	6.5 ± 2.24	20
TopK-PH+refit	0.02005 ± 0.003906	6.5 ± 2.24	20
TopK-IL	0.01874 ± 0.003652	6.5 ± 2.24	20
Mag-IL	0.01406 ± 0.002394	9.95 ± 2.46	20
Diag-x	0.05501 ± 0.1056	4.8 ± 1.67	20
ES	0.01922 ± 0.003661	6.5 ± 2.24	20
BF	0.02268 ± 0.001917	4.45 ± 1.15	20
MS-BF	0.01922 ± 0.003661	6.5 ± 2.24	20
SA	0.01922 ± 0.003661	6.5 ± 2.24	20
Tabu	0.01922 ± 0.003661	6.5 ± 2.24	20
amino, $\mu = 0.75$			
ALS ($R = 20$)	$0.00974 \pm 6.144\times10^{-5}$	20 ± 0	20
TopK-PH	0.7192 ± 0.1751	5.15 ± 2.16	20
TopK-PH+refit	0.03182 ± 0.04044	5.15 ± 2.16	20
TopK-IL	0.02601 ± 0.02196	5.15 ± 2.16	20
Mag-IL	0.01647 ± 0.003477	8.3 ± 2.58	20

Method	Rel. error	$\hat{R}$	n
Diag-x	0.05468 ± 0.1057	4.9 ± 1.71	20
ES	0.02153 ± 0.003372	5.15 ± 2.16	20
BF	0.02344 ± 0.001627	4 ± 0.858	20
MS-BF	0.02138 ± 0.003346	5.25 ± 2.15	20
SA	0.02153 ± 0.003372	5.15 ± 2.16	20
Tabu	0.02153 ± 0.003372	5.15 ± 2.16	20
covid, $\mu = 0.0001$			
ALS ($R = 20$)	$0.2183 \pm 8.366\times10^{-4}$	20 ± 0	20
TopK-PH	$0.2183 \pm 8.366\times10^{-4}$	20 ± 0	20
TopK-PH+refit	$0.2178 \pm 8.113\times10^{-4}$	20 ± 0	20
TopK-IL	$0.2183 \pm 8.366\times10^{-4}$	20 ± 0	20
Mag-IL	$0.2183 \pm 8.366\times10^{-4}$	20 ± 0	20
Diag-x	0.3893 ± 0.04901	6.1 ± 2.13	20
ES	$0.2183 \pm 8.366\times10^{-4}$	20 ± 0	20
BF	0.3057 ± 0.06943	11.8 ± 5.41	20
MS-BF	$0.2183 \pm 8.366\times10^{-4}$	20 ± 0	20
SA	$0.2183 \pm 8.366\times10^{-4}$	20 ± 0	20
Tabu	$0.2183 \pm 8.366\times10^{-4}$	20 ± 0	20
covid, $\mu = 0.65$			
ALS ($R = 20$)	$0.2183 \pm 8.366\times10^{-4}$	20 ± 0	20
TopK-PH	0.9506 ± 0.188	5.85 ± 2.3	20
TopK-PH+refit	0.3988 ± 0.05034	5.85 ± 2.3	20
TopK-IL	0.3967 ± 0.05016	5.85 ± 2.3	20
Mag-IL	0.2822 ± 0.03157	13.1 ± 2.87	20
Diag-x	0.4495 ± 0.02919	3.65 ± 0.875	20
ES	0.3958 ± 0.05077	5.85 ± 2.3	20
BF	0.4426 ± 0.05157	4 ± 1.56	20
MS-BF	0.3958 ± 0.05077	5.85 ± 2.3	20
SA	0.3958 ± 0.05077	5.85 ± 2.3	20
Tabu	0.3958 ± 0.05077	5.85 ± 2.3	20
covid, $\mu = 0.9$			
ALS ($R = 20$)	$0.2183 \pm 8.366\times10^{-4}$	20 ± 0	20
TopK-PH	1.035 ± 0.2241	4.35 ± 1.18	20
TopK-PH+refit	0.4325 ± 0.03349	4.35 ± 1.18	20
TopK-IL	0.4295 ± 0.03498	4.35 ± 1.18	20
Mag-IL	0.3376 ± 0.0306	8.75 ± 2.29	20
Diag-x	0.4741 ± 0.03688	2.95 ± 0.945	20
ES	0.4292 ± 0.03492	4.35 ± 1.18	20
BF	0.465 ± 0.04715	3.3 ± 1.22	20
MS-BF	0.4292 ± 0.03492	4.35 ± 1.18	20
SA	0.4292 ± 0.03492	4.35 ± 1.18	20
Tabu	0.4292 ± 0.03492	4.35 ± 1.18	20
kinetic, $\mu = 0.005$			
ALS ($R = 20$)	$0.02378 \pm 1.758\times10^{-4}$	20 ± 0	20
TopK-PH	0.05551 ± 0.02975	14.9 ± 2.21	20
TopK-PH+refit	$0.02468 \pm 6.430\times10^{-4}$	14.9 ± 2.21	20
TopK-IL	$0.02497 \pm 3.792\times10^{-4}$	14.9 ± 2.21	20
Mag-IL	$0.02488 \pm 3.208\times10^{-4}$	15.4 ± 1.96	20
Diag-x	0.02617 ± 0.001157	8.6 ± 1.93	20
ES	$0.02502 \pm 3.803\times10^{-4}$	14.9 ± 2.21	20
BF	0.02678 ± 0.002099	8 ± 1.92	20
MS-BF	$0.02502 \pm 3.803\times10^{-4}$	14.9 ± 2.21	20
SA	$0.02502 \pm 3.803\times10^{-4}$	14.9 ± 2.21	20
Tabu	$0.02506 \pm 3.705\times10^{-4}$	14.6 ± 2.54	20
kinetic, $\mu = 0.1$			

Method	Rel. error	$\hat{R}$	n
ALS ($R = 20$)	$0.02378 \pm 1.758\times10^{-4}$	20 ± 0	20
TopK-PH	0.2764 ± 0.116	5.95 ± 1.57	20
TopK-PH+refit	0.03017 ± 0.005111	5.95 ± 1.57	20
TopK-IL	0.02968 ± 0.005091	5.95 ± 1.57	20
Mag-IL	0.02607 ± 0.001147	9.1 ± 1.59	20
Diag-x	0.02629 ± 0.001082	8.25 ± 1.71	20
ES	0.02959 ± 0.005008	5.95 ± 1.57	20
BF	0.03457 ± 0.008522	4.65 ± 1.46	20
MS-BF	0.02959 ± 0.005008	5.95 ± 1.57	20
SA	0.02959 ± 0.005008	5.95 ± 1.57	20
Tabu	0.02886 ± 0.003891	6.1 ± 1.41	20
kinetic, $\mu = 0.8$			
ALS ($R = 20$)	$0.02378 \pm 1.758\times10^{-4}$	20 ± 0	20
TopK-PH	0.436 ± 0.1399	2.85 ± 0.875	20
TopK-PH+refit	0.05037 ± 0.02048	2.85 ± 0.875	20
TopK-IL	0.04996 ± 0.02063	2.85 ± 0.875	20
Mag-IL	0.05113 ± 0.02059	2.9 ± 1.29	20
Diag-x	0.02896 ± 0.004232	6.2 ± 1.44	20
ES	0.04984 ± 0.02072	2.85 ± 0.875	20
BF	0.05176 ± 0.02048	2.7 ± 0.865	20
MS-BF	0.04984 ± 0.02072	2.85 ± 0.875	20
SA	0.04984 ± 0.02072	2.85 ± 0.875	20
Tabu	0.04984 ± 0.02072	2.85 ± 0.875	20

S3 COVID-19 rows from $R_{\max} = 20$ (manuscript Table 10)

Table S3.

Method (setting)	$\hat{R}$	Rel. error
ALS ($R = 20$)	20	$0.2183 \pm 8.366\times10^{-4}$
ALS-LR (ES) ($\mu = 10^{-4}$)	20 ± 0	$0.2183 \pm 8.366\times10^{-4}$
ALS w/ Top-K ($K = 6$, $R_{\max} = 20$)	6	0.936 ± 0.1675
ALS w/ Top-K ($K = 8$, $R_{\max} = 20$)	8	0.8398 ± 0.1538
ALS w/ Top-K ($K = 10$, $R_{\max} = 20$)	10	0.7353 ± 0.1389
ALS w/ Top-K + refit ($K = 6$, $R_{\max} = 20$)	6	0.3884 ± 0.003542
ALS w/ Top-K + refit ($K = 8$, $R_{\max} = 20$)	8	0.35 ± 0.004017
ALS w/ Top-K + refit ($K = 10$, $R_{\max} = 20$)	10	0.3157 ± 0.002787

S4 Exact-SNR grid and SNR-matched collinearity (manuscript Tables 7 and 8)

Table S4. Every design at $\mu \in \{0.1, 0.3, 0.5\}$.

Method	Rel. error	$\hat{R}$	Excess over $\varepsilon^\star$	$\hat{R} = R_{\text{true}}$	n
rho0.0_snr+0dB, $\mu = 0.1$					
ALS ($R = 20$)	0.7 ± 0.00115	20 ± 0	-1.0%	0%	20
TopK-PH	0.7074 ± 0.002214	7 ± 0.973	0.0%	35%	20
TopK-PH+refit	0.7058 ± 0.001285	7 ± 0.973	-0.2%	35%	20
TopK-IL	0.7058 ± 0.001286	7 ± 0.973	-0.2%	35%	20
Mag-IL	0.7058 ± 0.0013	7.05 ± 0.945	-0.2%	30%	20
Diag-x	0.7054 ± 0.001357	8 ± 1.12	-0.2%	10%	20
ES	0.7058 ± 0.001286	7 ± 0.973	-0.2%	35%	20

Method	Rel. error	$\hat{R}$	Excess over $\varepsilon^\star$	$\hat{R} = R_{\text{true}}$	n
BF	0.7058 ± 0.001286	7 ± 0.973	-0.2%	35%	20
MS-BF	0.7058 ± 0.001286	7 ± 0.973	-0.2%	35%	20
SA	0.7058 ± 0.001286	7 ± 0.973	-0.2%	35%	20
Tabu	0.7058 ± 0.001286	7 ± 0.973	-0.2%	35%	20
rho0.0_snr+0dB, $\mu = 0.3$					
ALS ($R = 20$)	0.7 ± 0.00115	20 ± 0	-1.0%	0%	20
TopK-PH	0.7148 ± 0.009734	6.15 ± 0.366	1.1%	85%	20
TopK-PH+refit	0.706 ± 0.001218	6.15 ± 0.366	-0.2%	85%	20
TopK-IL	0.706 ± 0.001218	6.15 ± 0.366	-0.2%	85%	20
Mag-IL	0.706 ± 0.001218	6.15 ± 0.366	-0.2%	85%	20
Diag-x	0.7055 ± 0.001374	7.75 ± 1.12	-0.2%	15%	20
ES	0.706 ± 0.001218	6.15 ± 0.366	-0.2%	85%	20
BF	0.706 ± 0.001218	6.15 ± 0.366	-0.2%	85%	20
MS-BF	0.706 ± 0.001218	6.15 ± 0.366	-0.2%	85%	20
SA	0.706 ± 0.001218	6.15 ± 0.366	-0.2%	85%	20
Tabu	0.706 ± 0.001218	6.15 ± 0.366	-0.2%	85%	20
rho0.0_snr+0dB, $\mu = 0.5$					
ALS ($R = 20$)	0.7 ± 0.00115	20 ± 0	-1.0%	0%	20
TopK-PH	0.7167 ± 0.009699	6 ± 0	1.4%	100%	20
TopK-PH+refit	0.7061 ± 0.001169	6 ± 0	-0.1%	100%	20
TopK-IL	0.7061 ± 0.001169	6 ± 0	-0.1%	100%	20
Mag-IL	0.7287 ± 0.03362	5.6 ± 0.598	3.0%	65%	20
Diag-x	0.7057 ± 0.001324	7.1 ± 0.852	-0.2%	25%	20
ES	0.7061 ± 0.001169	6 ± 0	-0.1%	100%	20
BF	0.7061 ± 0.001169	6 ± 0	-0.1%	100%	20
MS-BF	0.7061 ± 0.001169	6 ± 0	-0.1%	100%	20
SA	0.7061 ± 0.001169	6 ± 0	-0.1%	100%	20
Tabu	0.7061 ± 0.001169	6 ± 0	-0.1%	100%	20
rho0.0_snr+10dB, $\mu = 0.1$					
ALS ($R = 20$)	$0.2987 \pm 3.337\times10^{-4}$	20 ± 0	-0.9%	0%	20
TopK-PH	0.3091 ± 0.01436	9.65 ± 1.53	2.5%	5%	20
TopK-PH+refit	$0.3006 \pm 3.653\times10^{-4}$	9.65 ± 1.53	-0.3%	5%	20
TopK-IL	$0.3006 \pm 3.612\times10^{-4}$	9.65 ± 1.53	-0.3%	5%	20
Mag-IL	$0.3006 \pm 3.612\times10^{-4}$	9.65 ± 1.53	-0.3%	5%	20
Diag-x	$0.3005 \pm 3.450\times10^{-4}$	10.6 ± 1.61	-0.3%	0%	20
ES	$0.3006 \pm 3.612\times10^{-4}$	9.65 ± 1.53	-0.3%	5%	20
BF	$0.3006 \pm 3.612\times10^{-4}$	9.65 ± 1.53	-0.3%	5%	20
MS-BF	$0.3006 \pm 3.612\times10^{-4}$	9.65 ± 1.53	-0.3%	5%	20
SA	$0.3006 \pm 3.612\times10^{-4}$	9.65 ± 1.53	-0.3%	5%	20
Tabu	$0.3006 \pm 3.612\times10^{-4}$	9.65 ± 1.53	-0.3%	5%	20
rho0.0_snr+10dB, $\mu = 0.3$					
ALS ($R = 20$)	$0.2987 \pm 3.337\times10^{-4}$	20 ± 0	-0.9%	0%	20
TopK-PH	0.334 ± 0.02719	8.15 ± 1.42	10.8%	10%	20
TopK-PH+refit	$0.3008 \pm 3.460\times10^{-4}$	8.15 ± 1.42	-0.2%	10%	20
TopK-IL	$0.3008 \pm 3.434\times10^{-4}$	8.15 ± 1.42	-0.2%	10%	20
Mag-IL	$0.3008 \pm 3.455\times10^{-4}$	8 ± 1.45	-0.2%	10%	20
Diag-x	$0.3005 \pm 3.509\times10^{-4}$	10.2 ± 1.58	-0.3%	5%	20
ES	$0.3008 \pm 3.431\times10^{-4}$	8.15 ± 1.42	-0.2%	10%	20
BF	$0.3008 \pm 3.354\times10^{-4}$	8 ± 1.26	-0.2%	10%	20
MS-BF	$0.3008 \pm 3.431\times10^{-4}$	8.15 ± 1.42	-0.2%	10%	20
SA	$0.3008 \pm 3.431\times10^{-4}$	8.15 ± 1.42	-0.2%	10%	20
Tabu	$0.3008 \pm 3.431\times10^{-4}$	8.15 ± 1.42	-0.2%	10%	20
rho0.0_snr+10dB, $\mu = 0.5$					
ALS ($R = 20$)	$0.2987 \pm 3.337\times10^{-4}$	20 ± 0	-0.9%	0%	20
TopK-PH	0.3808 ± 0.05569	6.9 ± 1.02	26.3%	45%	20

Method	Rel. error	$\hat{R}$	Excess over $\varepsilon^\star$	$\hat{R} = R_{\text{true}}$	n
TopK-PH+refit	$0.301 \pm 3.013\times10^{-4}$	6.9 ± 1.02	-0.2%	45%	20
TopK-IL	$0.301 \pm 3.037\times10^{-4}$	6.9 ± 1.02	-0.2%	45%	20
Mag-IL	0.3201 ± 0.05888	6.55 ± 0.999	6.2%	45%	20
Diag-x	$0.3006 \pm 3.578\times10^{-4}$	9.5 ± 1.43	-0.3%	5%	20
ES	$0.301 \pm 3.037\times10^{-4}$	6.9 ± 1.02	-0.2%	45%	20
BF	$0.301 \pm 3.037\times10^{-4}$	6.9 ± 1.02	-0.2%	45%	20
MS-BF	$0.301 \pm 3.037\times10^{-4}$	6.9 ± 1.02	-0.2%	45%	20
SA	$0.301 \pm 3.037\times10^{-4}$	6.9 ± 1.02	-0.2%	45%	20
Tabu	$0.301 \pm 3.037\times10^{-4}$	6.9 ± 1.02	-0.2%	45%	20
rho0.3_snr+0dB, $\mu = 0.1$					
ALS ($R = 20$)	$0.6996 \pm 9.748\times10^{-4}$	20 ± 0	-1.0%	0%	20
TopK-PH	0.7073 ± 0.00377	6.95 ± 0.826	0.1%	30%	20
TopK-PH+refit	$0.7055 \pm 9.687\times10^{-4}$	6.95 ± 0.826	-0.2%	30%	20
TopK-IL	$0.7055 \pm 9.664\times10^{-4}$	6.95 ± 0.826	-0.2%	30%	20
Mag-IL	$0.7055 \pm 9.664\times10^{-4}$	6.95 ± 0.826	-0.2%	30%	20
Diag-x	$0.7053 \pm 9.781\times10^{-4}$	7.35 ± 0.988	-0.2%	20%	20
ES	$0.7055 \pm 9.664\times10^{-4}$	6.95 ± 0.826	-0.2%	30%	20
BF	$0.7055 \pm 9.199\times10^{-4}$	6.85 ± 0.671	-0.2%	30%	20
MS-BF	$0.7055 \pm 9.664\times10^{-4}$	6.95 ± 0.826	-0.2%	30%	20
SA	$0.7055 \pm 9.664\times10^{-4}$	6.95 ± 0.826	-0.2%	30%	20
Tabu	$0.7055 \pm 9.664\times10^{-4}$	6.95 ± 0.826	-0.2%	30%	20
rho0.3_snr+0dB, $\mu = 0.3$					
ALS ($R = 20$)	$0.6996 \pm 9.748\times10^{-4}$	20 ± 0	-1.0%	0%	20
TopK-PH	0.7154 ± 0.01745	6.3 ± 0.47	1.2%	70%	20
TopK-PH+refit	0.7057 ± 0.001031	6.3 ± 0.47	-0.2%	70%	20
TopK-IL	0.7057 ± 0.001013	6.3 ± 0.47	-0.2%	70%	20
Mag-IL	$0.7058 \pm 9.411\times10^{-4}$	6.15 ± 0.366	-0.2%	85%	20
Diag-x	$0.7055 \pm 9.638\times10^{-4}$	7.05 ± 0.759	-0.2%	25%	20
ES	0.7057 ± 0.001008	6.3 ± 0.47	-0.2%	70%	20
BF	0.7057 ± 0.001008	6.3 ± 0.47	-0.2%	70%	20
MS-BF	0.7057 ± 0.001008	6.3 ± 0.47	-0.2%	70%	20
SA	0.7057 ± 0.001008	6.3 ± 0.47	-0.2%	70%	20
Tabu	0.7057 ± 0.001008	6.3 ± 0.47	-0.2%	70%	20
rho0.3_snr+0dB, $\mu = 0.5$					
ALS ($R = 20$)	$0.6996 \pm 9.748\times10^{-4}$	20 ± 0	-1.0%	0%	20
TopK-PH	0.719 ± 0.02536	6.05 ± 0.224	1.7%	95%	20
TopK-PH+refit	$0.7058 \pm 9.482\times10^{-4}$	6.05 ± 0.224	-0.1%	95%	20
TopK-IL	$0.7058 \pm 9.482\times10^{-4}$	6.05 ± 0.224	-0.1%	95%	20
Mag-IL	0.7205 ± 0.02754	5.7 ± 0.571	1.9%	75%	20
Diag-x	$0.7055 \pm 9.908\times10^{-4}$	6.9 ± 0.852	-0.2%	35%	20
ES	$0.7058 \pm 9.482\times10^{-4}$	6.05 ± 0.224	-0.1%	95%	20
BF	$0.7058 \pm 9.482\times10^{-4}$	6.05 ± 0.224	-0.1%	95%	20
MS-BF	$0.7058 \pm 9.482\times10^{-4}$	6.05 ± 0.224	-0.1%	95%	20
SA	$0.7058 \pm 9.482\times10^{-4}$	6.05 ± 0.224	-0.1%	95%	20
Tabu	$0.7058 \pm 9.482\times10^{-4}$	6.05 ± 0.224	-0.1%	95%	20
rho0.3_snr+10dB, $\mu = 0.1$					
ALS ($R = 20$)	$0.2987 \pm 2.797\times10^{-4}$	20 ± 0	-0.9%	0%	20
TopK-PH	0.31 ± 0.0104	9.7 ± 1.26	2.8%	0%	20
TopK-PH+refit	$0.3005 \pm 2.285\times10^{-4}$	9.7 ± 1.26	-0.3%	0%	20
TopK-IL	$0.3005 \pm 2.465\times10^{-4}$	9.7 ± 1.26	-0.3%	0%	20
Mag-IL	$0.3005 \pm 2.599\times10^{-4}$	9.65 ± 1.27	-0.3%	0%	20
Diag-x	$0.3004 \pm 2.250\times10^{-4}$	10.3 ± 1.5	-0.3%	0%	20
ES	$0.3005 \pm 2.465\times10^{-4}$	9.7 ± 1.26	-0.3%	0%	20
BF	$0.3005 \pm 2.467\times10^{-4}$	9.6 ± 1.23	-0.3%	0%	20
MS-BF	$0.3005 \pm 2.465\times10^{-4}$	9.7 ± 1.26	-0.3%	0%	20

Method	Rel. error	$\hat{R}$	Excess over $\varepsilon^\star$	$\hat{R} = R_{\text{true}}$	n
SA	$0.3005 \pm 2.465\times10^{-4}$	9.7 ± 1.26	-0.3%	0%	20
Tabu	$0.3005 \pm 2.465\times10^{-4}$	9.7 ± 1.26	-0.3%	0%	20
rho0.3_snr+10dB, $\mu = 0.3$					
ALS ($R = 20$)	$0.2987 \pm 2.797\times10^{-4}$	20 ± 0	-0.9%	0%	20
TopK-PH	0.3438 ± 0.02832	7.95 ± 0.826	14.0%	0%	20
TopK-PH+refit	$0.3008 \pm 2.814\times10^{-4}$	7.95 ± 0.826	-0.2%	0%	20
TopK-IL	$0.3008 \pm 2.885\times10^{-4}$	7.95 ± 0.826	-0.2%	0%	20
Mag-IL	$0.3008 \pm 2.817\times10^{-4}$	7.7 ± 0.865	-0.2%	5%	20
Diag-x	$0.3005 \pm 2.028\times10^{-4}$	10.2 ± 1.36	-0.3%	0%	20
ES	$0.3008 \pm 2.824\times10^{-4}$	7.95 ± 0.826	-0.2%	0%	20
BF	$0.3008 \pm 2.824\times10^{-4}$	7.95 ± 0.826	-0.2%	0%	20
MS-BF	$0.3008 \pm 2.824\times10^{-4}$	7.95 ± 0.826	-0.2%	0%	20
SA	$0.3008 \pm 2.824\times10^{-4}$	7.95 ± 0.826	-0.2%	0%	20
Tabu	$0.3008 \pm 2.824\times10^{-4}$	7.95 ± 0.826	-0.2%	0%	20
rho0.3_snr+10dB, $\mu = 0.5$					
ALS ($R = 20$)	$0.2987 \pm 2.797\times10^{-4}$	20 ± 0	-0.9%	0%	20
TopK-PH	0.3823 ± 0.03967	6.95 ± 0.686	26.8%	25%	20
TopK-PH+refit	$0.3009 \pm 2.703\times10^{-4}$	6.95 ± 0.686	-0.2%	25%	20
TopK-IL	$0.3009 \pm 2.725\times10^{-4}$	6.95 ± 0.686	-0.2%	25%	20
Mag-IL	$0.3009 \pm 2.883\times10^{-4}$	6.7 ± 0.733	-0.2%	45%	20
Diag-x	$0.3005 \pm 1.820\times10^{-4}$	9.6 ± 1.27	-0.3%	0%	20
ES	$0.3009 \pm 2.713\times10^{-4}$	6.95 ± 0.686	-0.2%	25%	20
BF	$0.3009 \pm 2.713\times10^{-4}$	6.95 ± 0.686	-0.2%	25%	20
MS-BF	$0.3009 \pm 2.713\times10^{-4}$	6.95 ± 0.686	-0.2%	25%	20
SA	$0.3009 \pm 2.713\times10^{-4}$	6.95 ± 0.686	-0.2%	25%	20
Tabu	$0.3009 \pm 2.713\times10^{-4}$	6.95 ± 0.686	-0.2%	25%	20
rho0.5_snr+0dB, $\mu = 0.1$					
ALS ($R = 20$)	$0.6996 \pm 9.141\times10^{-4}$	20 ± 0	-1.0%	0%	20
TopK-PH	0.7072 ± 0.006226	6.9 ± 0.852	0.1%	35%	20
TopK-PH+refit	$0.7054 \pm 9.413\times10^{-4}$	6.9 ± 0.852	-0.2%	35%	20
TopK-IL	$0.7055 \pm 9.375\times10^{-4}$	6.9 ± 0.852	-0.2%	35%	20
Mag-IL	$0.7055 \pm 9.522\times10^{-4}$	6.85 ± 0.813	-0.2%	35%	20
Diag-x	$0.7054 \pm 9.926\times10^{-4}$	7.1 ± 0.788	-0.2%	25%	20
ES	$0.7055 \pm 9.375\times10^{-4}$	6.9 ± 0.852	-0.2%	35%	20
BF	$0.7055 \pm 9.375\times10^{-4}$	6.9 ± 0.852	-0.2%	35%	20
MS-BF	$0.7055 \pm 9.375\times10^{-4}$	6.9 ± 0.852	-0.2%	35%	20
SA	$0.7055 \pm 9.375\times10^{-4}$	6.9 ± 0.852	-0.2%	35%	20
Tabu	$0.7055 \pm 9.375\times10^{-4}$	6.9 ± 0.852	-0.2%	35%	20
rho0.5_snr+0dB, $\mu = 0.3$					
ALS ($R = 20$)	$0.6996 \pm 9.141\times10^{-4}$	20 ± 0	-1.0%	0%	20
TopK-PH	0.7144 ± 0.01302	6.15 ± 0.366	1.1%	85%	20
TopK-PH+refit	$0.7057 \pm 9.334\times10^{-4}$	6.15 ± 0.366	-0.2%	85%	20
TopK-IL	$0.7057 \pm 9.333\times10^{-4}$	6.15 ± 0.366	-0.2%	85%	20
Mag-IL	$0.7057 \pm 9.316\times10^{-4}$	6.05 ± 0.224	-0.2%	95%	20
Diag-x	$0.7054 \pm 9.334\times10^{-4}$	6.95 ± 0.759	-0.2%	30%	20
ES	$0.7057 \pm 9.333\times10^{-4}$	6.15 ± 0.366	-0.2%	85%	20
BF	$0.7057 \pm 9.333\times10^{-4}$	6.15 ± 0.366	-0.2%	85%	20
MS-BF	$0.7057 \pm 9.333\times10^{-4}$	6.15 ± 0.366	-0.2%	85%	20
SA	$0.7057 \pm 9.333\times10^{-4}$	6.15 ± 0.366	-0.2%	85%	20
Tabu	$0.7057 \pm 9.333\times10^{-4}$	6.15 ± 0.366	-0.2%	85%	20
rho0.5_snr+0dB, $\mu = 0.5$					
ALS ($R = 20$)	$0.6996 \pm 9.141\times10^{-4}$	20 ± 0	-1.0%	0%	20
TopK-PH	0.7158 ± 0.01277	6 ± 0	1.3%	100%	20
TopK-PH+refit	$0.7058 \pm 9.329\times10^{-4}$	6 ± 0	-0.1%	100%	20
TopK-IL	$0.7058 \pm 9.329\times10^{-4}$	6 ± 0	-0.1%	100%	20

Method	Rel. error	$\hat{R}$	Excess over $\varepsilon^\star$	$\hat{R} = R_{\text{true}}$	n
Mag-IL	0.7208 ± 0.0208	5.45 ± 0.686	2.0%	55%	20
Diag-x	$0.7055 \pm 8.895\times10^{-4}$	6.75 ± 0.639	-0.2%	35%	20
ES	$0.7058 \pm 9.329\times10^{-4}$	6 ± 0	-0.1%	100%	20
BF	$0.7058 \pm 9.329\times10^{-4}$	6 ± 0	-0.1%	100%	20
MS-BF	$0.7058 \pm 9.329\times10^{-4}$	6 ± 0	-0.1%	100%	20
SA	$0.7058 \pm 9.329\times10^{-4}$	6 ± 0	-0.1%	100%	20
Tabu	$0.7058 \pm 9.329\times10^{-4}$	6 ± 0	-0.1%	100%	20
rho0.5_snr+10dB, $\mu = 0.1$					
ALS ($R = 20$)	$0.2987 \pm 2.874\times10^{-4}$	20 ± 0	-0.9%	0%	20
TopK-PH	0.3153 ± 0.01544	9.8 ± 2.26	4.6%	0%	20
TopK-PH+refit	$0.3005 \pm 3.241\times10^{-4}$	9.8 ± 2.26	-0.3%	0%	20
TopK-IL	$0.3005 \pm 3.267\times10^{-4}$	9.8 ± 2.26	-0.3%	0%	20
Mag-IL	$0.3005 \pm 3.322\times10^{-4}$	9.85 ± 2.46	-0.3%	0%	20
Diag-x	$0.3003 \pm 2.907\times10^{-4}$	10.9 ± 1.82	-0.4%	0%	20
ES	$0.3005 \pm 3.275\times10^{-4}$	9.8 ± 2.26	-0.3%	0%	20
BF	$0.3005 \pm 3.160\times10^{-4}$	9.7 ± 2.05	-0.3%	0%	20
MS-BF	$0.3005 \pm 3.275\times10^{-4}$	9.8 ± 2.26	-0.3%	0%	20
SA	$0.3005 \pm 3.275\times10^{-4}$	9.8 ± 2.26	-0.3%	0%	20
Tabu	$0.3005 \pm 3.275\times10^{-4}$	9.8 ± 2.26	-0.3%	0%	20
rho0.5_snr+10dB, $\mu = 0.3$					
ALS ($R = 20$)	$0.2987 \pm 2.874\times10^{-4}$	20 ± 0	-0.9%	0%	20
TopK-PH	0.3645 ± 0.03444	7.5 ± 1.24	20.9%	30%	20
TopK-PH+refit	$0.3008 \pm 2.929\times10^{-4}$	7.5 ± 1.24	-0.2%	30%	20
TopK-IL	$0.3008 \pm 2.880\times10^{-4}$	7.5 ± 1.24	-0.2%	30%	20
Mag-IL	$0.3009 \pm 2.567\times10^{-4}$	7.05 ± 1.1	-0.2%	45%	20
Diag-x	$0.3004 \pm 2.754\times10^{-4}$	10.3 ± 2.11	-0.3%	0%	20
ES	$0.3008 \pm 2.879\times10^{-4}$	7.5 ± 1.24	-0.2%	30%	20
BF	$0.3008 \pm 2.879\times10^{-4}$	7.5 ± 1.24	-0.2%	30%	20
MS-BF	$0.3008 \pm 2.879\times10^{-4}$	7.5 ± 1.24	-0.2%	30%	20
SA	$0.3008 \pm 2.879\times10^{-4}$	7.5 ± 1.24	-0.2%	30%	20
Tabu	$0.3008 \pm 2.879\times10^{-4}$	7.5 ± 1.24	-0.2%	30%	20
rho0.5_snr+10dB, $\mu = 0.5$					
ALS ($R = 20$)	$0.2987 \pm 2.874\times10^{-4}$	20 ± 0	-0.9%	0%	20
TopK-PH	0.4005 ± 0.04978	6.55 ± 0.826	32.9%	60%	20
TopK-PH+refit	$0.3009 \pm 2.355\times10^{-4}$	6.55 ± 0.826	-0.2%	60%	20
TopK-IL	$0.3009 \pm 2.354\times10^{-4}$	6.55 ± 0.826	-0.2%	60%	20
Mag-IL	0.3096 ± 0.02661	6.05 ± 0.51	2.7%	75%	20
Diag-x	$0.3005 \pm 2.911\times10^{-4}$	9.85 ± 1.87	-0.3%	0%	20
ES	$0.3009 \pm 2.332\times10^{-4}$	6.55 ± 0.826	-0.2%	60%	20
BF	$0.3009 \pm 2.332\times10^{-4}$	6.55 ± 0.826	-0.2%	60%	20
MS-BF	$0.3009 \pm 2.332\times10^{-4}$	6.55 ± 0.826	-0.2%	60%	20
SA	$0.3009 \pm 2.332\times10^{-4}$	6.55 ± 0.826	-0.2%	60%	20
Tabu	$0.3009 \pm 2.332\times10^{-4}$	6.55 ± 0.826	-0.2%	60%	20
rho0.7_snr+0dB, $\mu = 0.1$					
ALS ($R = 20$)	$0.6995 \pm 8.527\times10^{-4}$	20 ± 0	-1.0%	0%	20
TopK-PH	0.7082 ± 0.004964	6.3 ± 0.571	0.2%	75%	20
TopK-PH+refit	0.7057 ± 0.001007	6.3 ± 0.571	-0.2%	75%	20
TopK-IL	$0.7057 \pm 9.962\times10^{-4}$	6.3 ± 0.571	-0.2%	75%	20
Mag-IL	$0.7057 \pm 9.736\times10^{-4}$	6.25 ± 0.444	-0.2%	75%	20
Diag-x	0.7054 ± 0.00106	7.1 ± 0.718	-0.2%	20%	20
ES	$0.7057 \pm 9.962\times10^{-4}$	6.3 ± 0.571	-0.2%	75%	20
BF	$0.7057 \pm 9.962\times10^{-4}$	6.3 ± 0.571	-0.2%	75%	20
MS-BF	$0.7057 \pm 9.962\times10^{-4}$	6.3 ± 0.571	-0.2%	75%	20
SA	$0.7057 \pm 9.962\times10^{-4}$	6.3 ± 0.571	-0.2%	75%	20
Tabu	$0.7057 \pm 9.962\times10^{-4}$	6.3 ± 0.571	-0.2%	75%	20

Method	Rel. error	$\hat{R}$	Excess over $\varepsilon^\star$	$\hat{R} = R_{\text{true}}$	n
rho0.7_snr+0dB, $\mu = 0.3$					
ALS ($R = 20$)	$0.6995 \pm 8.527\times10^{-4}$	20 ± 0	-1.0%	0%	20
TopK-PH	0.7097 ± 0.00526	6 ± 0	0.4%	100%	20
TopK-PH+refit	$0.7058 \pm 9.325\times10^{-4}$	6 ± 0	-0.1%	100%	20
TopK-IL	$0.7058 \pm 9.325\times10^{-4}$	6 ± 0	-0.1%	100%	20
Mag-IL	0.7093 ± 0.004169	5.5 ± 0.607	0.3%	55%	20
Diag-x	0.7055 ± 0.00102	6.75 ± 0.639	-0.2%	35%	20
ES	$0.7058 \pm 9.325\times10^{-4}$	6 ± 0	-0.1%	100%	20
BF	$0.7058 \pm 9.325\times10^{-4}$	6 ± 0	-0.1%	100%	20
MS-BF	$0.7058 \pm 9.325\times10^{-4}$	6 ± 0	-0.1%	100%	20
SA	$0.7058 \pm 9.325\times10^{-4}$	6 ± 0	-0.1%	100%	20
Tabu	$0.7058 \pm 9.325\times10^{-4}$	6 ± 0	-0.1%	100%	20
rho0.7_snr+0dB, $\mu = 0.5$					
ALS ($R = 20$)	$0.6995 \pm 8.527\times10^{-4}$	20 ± 0	-1.0%	0%	20
TopK-PH	0.7097 ± 0.00526	6 ± 0	0.4%	100%	20
TopK-PH+refit	$0.7058 \pm 9.325\times10^{-4}$	6 ± 0	-0.1%	100%	20
TopK-IL	$0.7058 \pm 9.325\times10^{-4}$	6 ± 0	-0.1%	100%	20
Mag-IL	0.7215 ± 0.01104	4.05 ± 1.19	2.1%	10%	20
Diag-x	0.7056 ± 0.001027	6.65 ± 0.671	-0.2%	45%	20
ES	$0.7058 \pm 9.325\times10^{-4}$	6 ± 0	-0.1%	100%	20
BF	$0.7058 \pm 9.325\times10^{-4}$	6 ± 0	-0.1%	100%	20
MS-BF	$0.7058 \pm 9.325\times10^{-4}$	6 ± 0	-0.1%	100%	20
SA	$0.7058 \pm 9.325\times10^{-4}$	6 ± 0	-0.1%	100%	20
Tabu	$0.7058 \pm 9.325\times10^{-4}$	6 ± 0	-0.1%	100%	20
rho0.7_snr+10dB, $\mu = 0.1$					
ALS ($R = 20$)	$0.2986 \pm 2.347\times10^{-4}$	20 ± 0	-0.9%	0%	20
TopK-PH	0.3175 ± 0.01315	8.55 ± 1.23	5.3%	0%	20
TopK-PH+refit	$0.3007 \pm 2.670\times10^{-4}$	8.55 ± 1.23	-0.3%	0%	20
TopK-IL	$0.3006 \pm 2.656\times10^{-4}$	8.55 ± 1.23	-0.3%	0%	20
Mag-IL	$0.3007 \pm 2.699\times10^{-4}$	8.4 ± 1.43	-0.3%	0%	20
Diag-x	$0.3003 \pm 2.836\times10^{-4}$	10.9 ± 1.71	-0.4%	0%	20
ES	$0.3006 \pm 2.657\times10^{-4}$	8.55 ± 1.23	-0.3%	0%	20
BF	$0.3006 \pm 2.688\times10^{-4}$	8.6 ± 1.23	-0.3%	0%	20
MS-BF	$0.3006 \pm 2.657\times10^{-4}$	8.55 ± 1.23	-0.3%	0%	20
SA	$0.3006 \pm 2.657\times10^{-4}$	8.55 ± 1.23	-0.3%	0%	20
Tabu	$0.3006 \pm 2.657\times10^{-4}$	8.55 ± 1.23	-0.3%	0%	20
rho0.7_snr+10dB, $\mu = 0.3$					
ALS ($R = 20$)	$0.2986 \pm 2.347\times10^{-4}$	20 ± 0	-0.9%	0%	20
TopK-PH	0.3795 ± 0.06033	6.5 ± 0.607	25.9%	55%	20
TopK-PH+refit	$0.3009 \pm 2.364\times10^{-4}$	6.5 ± 0.607	-0.2%	55%	20
TopK-IL	$0.3009 \pm 2.392\times10^{-4}$	6.5 ± 0.607	-0.2%	55%	20
Mag-IL	0.3029 ± 0.006338	6.05 ± 0.51	0.5%	75%	20
Diag-x	$0.3004 \pm 2.739\times10^{-4}$	10.3 ± 1.42	-0.3%	0%	20
ES	$0.3009 \pm 2.382\times10^{-4}$	6.5 ± 0.607	-0.2%	55%	20
BF	$0.3009 \pm 2.382\times10^{-4}$	6.5 ± 0.607	-0.2%	55%	20
MS-BF	$0.3009 \pm 2.382\times10^{-4}$	6.5 ± 0.607	-0.2%	55%	20
SA	$0.3009 \pm 2.382\times10^{-4}$	6.5 ± 0.607	-0.2%	55%	20
Tabu	$0.3009 \pm 2.382\times10^{-4}$	6.5 ± 0.607	-0.2%	55%	20
rho0.7_snr+10dB, $\mu = 0.5$					
ALS ($R = 20$)	$0.2986 \pm 2.347\times10^{-4}$	20 ± 0	-0.9%	0%	20
TopK-PH	0.4074 ± 0.06312	5.95 ± 0.224	35.2%	95%	20
TopK-PH+refit	0.3026 ± 0.004866	5.95 ± 0.224	0.4%	95%	20
TopK-IL	0.3016 ± 0.002762	5.95 ± 0.224	0.1%	95%	20
Mag-IL	0.3347 ± 0.02964	4.75 ± 1.02	11.0%	20%	20
Diag-x	$0.3005 \pm 2.474\times10^{-4}$	9.85 ± 1.27	-0.3%	0%	20

Method	Rel. error	$\hat{R}$	Excess over $\varepsilon^\star$	$\hat{R} = R_{\text{true}}$	n
ES	0.3016 ± 0.002762	5.95 ± 0.224	0.1%	95%	20
BF	0.3016 ± 0.002762	5.95 ± 0.224	0.1%	95%	20
MS-BF	0.3016 ± 0.002762	5.95 ± 0.224	0.1%	95%	20
SA	0.3016 ± 0.002762	5.95 ± 0.224	0.1%	95%	20
Tabu	0.3016 ± 0.002762	5.95 ± 0.224	0.1%	95%	20
rho0.9_snr+0dB, $\mu = 0.1$					
ALS ($R = 20$)	$0.6994 \pm 9.513\times10^{-4}$	20 ± 0	-1.1%	0%	20
TopK-PH	0.7094 ± 0.004534	5.35 ± 0.813	0.4%	25%	20
TopK-PH+refit	$0.7061 \pm 9.550\times10^{-4}$	5.35 ± 0.813	-0.1%	25%	20
TopK-IL	$0.7061 \pm 9.485\times10^{-4}$	5.35 ± 0.813	-0.1%	25%	20
Mag-IL	$0.7063 \pm 9.838\times10^{-4}$	5.05 ± 1.19	-0.1%	20%	20
Diag-x	$0.7057 \pm 9.063\times10^{-4}$	6.25 ± 0.716	-0.2%	60%	20
ES	$0.7061 \pm 9.485\times10^{-4}$	5.35 ± 0.813	-0.1%	25%	20
BF	$0.7061 \pm 9.485\times10^{-4}$	5.35 ± 0.813	-0.1%	25%	20
MS-BF	$0.7061 \pm 9.485\times10^{-4}$	5.35 ± 0.813	-0.1%	25%	20
SA	$0.7061 \pm 9.485\times10^{-4}$	5.35 ± 0.813	-0.1%	25%	20
Tabu	$0.7061 \pm 9.485\times10^{-4}$	5.35 ± 0.813	-0.1%	25%	20
rho0.9_snr+0dB, $\mu = 0.3$					
ALS ($R = 20$)	$0.6994 \pm 9.513\times10^{-4}$	20 ± 0	-1.1%	0%	20
TopK-PH	0.7347 ± 0.01729	3.7 ± 0.979	3.9%	5%	20
TopK-PH+refit	0.7073 ± 0.001061	3.7 ± 0.979	0.1%	5%	20
TopK-IL	0.7073 ± 0.001054	3.7 ± 0.979	0.1%	5%	20
Mag-IL	0.7085 ± 0.001229	2.65 ± 0.813	0.2%	0%	20
Diag-x	$0.7057 \pm 8.881\times10^{-4}$	6.15 ± 0.671	-0.2%	55%	20
ES	0.7073 ± 0.001054	3.7 ± 0.979	0.1%	5%	20
BF	0.7073 ± 0.001054	3.7 ± 0.979	0.1%	5%	20
MS-BF	0.7073 ± 0.001054	3.7 ± 0.979	0.1%	5%	20
SA	0.7073 ± 0.001054	3.7 ± 0.979	0.1%	5%	20
Tabu	0.7073 ± 0.001054	3.7 ± 0.979	0.1%	5%	20
rho0.9_snr+0dB, $\mu = 0.5$					
ALS ($R = 20$)	$0.6994 \pm 9.513\times10^{-4}$	20 ± 0	-1.1%	0%	20
TopK-PH	0.7709 ± 0.0273	2.65 ± 0.587	9.1%	0%	20
TopK-PH+refit	0.7085 ± 0.001153	2.65 ± 0.587	0.2%	0%	20
TopK-IL	0.7085 ± 0.001149	2.65 ± 0.587	0.2%	0%	20
Mag-IL	0.7101 ± 0.002313	1.65 ± 0.671	0.5%	0%	20
Diag-x	$0.7059 \pm 9.171\times10^{-4}$	5.85 ± 0.875	-0.1%	40%	20
ES	0.7085 ± 0.001149	2.65 ± 0.587	0.2%	0%	20
BF	0.7085 ± 0.001149	2.65 ± 0.587	0.2%	0%	20
MS-BF	0.7085 ± 0.001149	2.65 ± 0.587	0.2%	0%	20
SA	0.7085 ± 0.001149	2.65 ± 0.587	0.2%	0%	20
Tabu	0.7085 ± 0.001149	2.65 ± 0.587	0.2%	0%	20
rho0.9_snr+10dB, $\mu = 0.1$					
ALS ($R = 20$)	$0.2985 \pm 2.494\times10^{-4}$	20 ± 0	-1.0%	0%	20
TopK-PH	0.3266 ± 0.02395	7.15 ± 0.813	8.3%	20%	20
TopK-PH+refit	$0.3009 \pm 2.453\times10^{-4}$	7.15 ± 0.813	-0.2%	20%	20
TopK-IL	$0.3008 \pm 2.436\times10^{-4}$	7.15 ± 0.813	-0.2%	20%	20
Mag-IL	$0.301 \pm 4.139\times10^{-4}$	6.85 ± 1.14	-0.1%	30%	20
Diag-x	$0.3005 \pm 2.643\times10^{-4}$	9.85 ± 1.31	-0.3%	0%	20
ES	$0.3008 \pm 2.429\times10^{-4}$	7.15 ± 0.813	-0.2%	20%	20
BF	$0.3008 \pm 2.429\times10^{-4}$	7.15 ± 0.813	-0.2%	20%	20
MS-BF	$0.3008 \pm 2.429\times10^{-4}$	7.15 ± 0.813	-0.2%	20%	20
SA	$0.3008 \pm 2.429\times10^{-4}$	7.15 ± 0.813	-0.2%	20%	20
Tabu	$0.3008 \pm 2.429\times10^{-4}$	7.15 ± 0.813	-0.2%	20%	20
rho0.9_snr+10dB, $\mu = 0.3$					
ALS ($R = 20$)	$0.2985 \pm 2.494\times10^{-4}$	20 ± 0	-1.0%	0%	20

Method	Rel. error	$\hat{R}$	Excess over $\varepsilon^\star$	$\hat{R} = R_{\text{true}}$	n
TopK-PH	0.3588 ± 0.03449	5.85 ± 0.366	19.0%	85%	20
TopK-PH+refit	$0.3012 \pm 5.964\times10^{-4}$	5.85 ± 0.366	-0.1%	85%	20
TopK-IL	$0.3012 \pm 5.939\times10^{-4}$	5.85 ± 0.366	-0.1%	85%	20
Mag-IL	0.3074 ± 0.003247	3.4 ± 0.995	2.0%	0%	20
Diag-x	$0.3005 \pm 2.481\times10^{-4}$	9.55 ± 1.28	-0.3%	0%	20
ES	$0.3012 \pm 5.938\times10^{-4}$	5.85 ± 0.366	-0.1%	85%	20
BF	$0.3012 \pm 5.938\times10^{-4}$	5.85 ± 0.366	-0.1%	85%	20
MS-BF	$0.3012 \pm 5.938\times10^{-4}$	5.85 ± 0.366	-0.1%	85%	20
SA	$0.3012 \pm 5.938\times10^{-4}$	5.85 ± 0.366	-0.1%	85%	20
Tabu	$0.3012 \pm 5.938\times10^{-4}$	5.85 ± 0.366	-0.1%	85%	20
rho0.9_snr+10dB, $\mu = 0.5$					
ALS ($R = 20$)	$0.2985 \pm 2.494\times10^{-4}$	20 ± 0	-1.0%	0%	20
TopK-PH	0.4307 ± 0.07139	4.8 ± 0.894	42.9%	25%	20
TopK-PH+refit	0.3029 ± 0.001296	4.8 ± 0.894	0.5%	25%	20
TopK-IL	0.3029 ± 0.001285	4.8 ± 0.894	0.5%	25%	20
Mag-IL	0.3135 ± 0.005592	2 ± 0.858	4.0%	0%	20
Diag-x	$0.3006 \pm 2.660\times10^{-4}$	9.2 ± 1.36	-0.3%	0%	20
ES	0.3029 ± 0.001283	4.8 ± 0.894	0.5%	25%	20
BF	0.3028 ± 0.001322	4.85 ± 0.933	0.4%	30%	20
MS-BF	0.3029 ± 0.001283	4.8 ± 0.894	0.5%	25%	20
SA	0.3029 ± 0.001283	4.8 ± 0.894	0.5%	25%	20
Tabu	0.3029 ± 0.001283	4.8 ± 0.894	0.5%	25%	20
snr+0dB_rho0.5, $\mu = 0.1$					
ALS ($R = 20$)	$0.6996 \pm 9.141\times10^{-4}$	20 ± 0	-1.0%	0%	20
TopK-PH	0.7072 ± 0.006226	6.9 ± 0.852	0.1%	35%	20
TopK-PH+refit	$0.7054 \pm 9.413\times10^{-4}$	6.9 ± 0.852	-0.2%	35%	20
TopK-IL	$0.7055 \pm 9.375\times10^{-4}$	6.9 ± 0.852	-0.2%	35%	20
Mag-IL	$0.7055 \pm 9.522\times10^{-4}$	6.85 ± 0.813	-0.2%	35%	20
Diag-x	$0.7054 \pm 9.926\times10^{-4}$	7.1 ± 0.788	-0.2%	25%	20
ES	$0.7055 \pm 9.375\times10^{-4}$	6.9 ± 0.852	-0.2%	35%	20
BF	$0.7055 \pm 9.375\times10^{-4}$	6.9 ± 0.852	-0.2%	35%	20
MS-BF	$0.7055 \pm 9.375\times10^{-4}$	6.9 ± 0.852	-0.2%	35%	20
SA	$0.7055 \pm 9.375\times10^{-4}$	6.9 ± 0.852	-0.2%	35%	20
Tabu	$0.7055 \pm 9.375\times10^{-4}$	6.9 ± 0.852	-0.2%	35%	20
snr+0dB_rho0.5, $\mu = 0.3$					
ALS ($R = 20$)	$0.6996 \pm 9.141\times10^{-4}$	20 ± 0	-1.0%	0%	20
TopK-PH	0.7144 ± 0.01302	6.15 ± 0.366	1.1%	85%	20
TopK-PH+refit	$0.7057 \pm 9.334\times10^{-4}$	6.15 ± 0.366	-0.2%	85%	20
TopK-IL	$0.7057 \pm 9.333\times10^{-4}$	6.15 ± 0.366	-0.2%	85%	20
Mag-IL	$0.7057 \pm 9.316\times10^{-4}$	6.05 ± 0.224	-0.2%	95%	20
Diag-x	$0.7054 \pm 9.334\times10^{-4}$	6.95 ± 0.759	-0.2%	30%	20
ES	$0.7057 \pm 9.333\times10^{-4}$	6.15 ± 0.366	-0.2%	85%	20
BF	$0.7057 \pm 9.333\times10^{-4}$	6.15 ± 0.366	-0.2%	85%	20
MS-BF	$0.7057 \pm 9.333\times10^{-4}$	6.15 ± 0.366	-0.2%	85%	20
SA	$0.7057 \pm 9.333\times10^{-4}$	6.15 ± 0.366	-0.2%	85%	20
Tabu	$0.7057 \pm 9.333\times10^{-4}$	6.15 ± 0.366	-0.2%	85%	20
snr+0dB_rho0.5, $\mu = 0.5$					
ALS ($R = 20$)	$0.6996 \pm 9.141\times10^{-4}$	20 ± 0	-1.0%	0%	20
TopK-PH	0.7158 ± 0.01277	6 ± 0	1.3%	100%	20
TopK-PH+refit	$0.7058 \pm 9.329\times10^{-4}$	6 ± 0	-0.1%	100%	20
TopK-IL	$0.7058 \pm 9.329\times10^{-4}$	6 ± 0	-0.1%	100%	20
Mag-IL	0.7208 ± 0.0208	5.45 ± 0.686	2.0%	55%	20
Diag-x	$0.7055 \pm 8.895\times10^{-4}$	6.75 ± 0.639	-0.2%	35%	20
ES	$0.7058 \pm 9.329\times10^{-4}$	6 ± 0	-0.1%	100%	20
BF	$0.7058 \pm 9.329\times10^{-4}$	6 ± 0	-0.1%	100%	20

Method	Rel. error	$\hat{R}$	Excess over $\varepsilon^\star$	$\hat{R} = R_{\text{true}}$	n
MS-BF	$0.7058 \pm 9.329\times10^{-4}$	6 ± 0	-0.1%	100%	20
SA	$0.7058 \pm 9.329\times10^{-4}$	6 ± 0	-0.1%	100%	20
Tabu	$0.7058 \pm 9.329\times10^{-4}$	6 ± 0	-0.1%	100%	20
snr+10dB_rho0.5, $\mu = 0.1$					
ALS ($R = 20$)	$0.2987 \pm 2.874\times10^{-4}$	20 ± 0	-0.9%	0%	20
TopK-PH	0.3153 ± 0.01544	9.8 ± 2.26	4.6%	0%	20
TopK-PH+refit	$0.3005 \pm 3.241\times10^{-4}$	9.8 ± 2.26	-0.3%	0%	20
TopK-IL	$0.3005 \pm 3.267\times10^{-4}$	9.8 ± 2.26	-0.3%	0%	20
Mag-IL	$0.3005 \pm 3.322\times10^{-4}$	9.85 ± 2.46	-0.3%	0%	20
Diag-x	$0.3003 \pm 2.907\times10^{-4}$	10.9 ± 1.82	-0.4%	0%	20
ES	$0.3005 \pm 3.275\times10^{-4}$	9.8 ± 2.26	-0.3%	0%	20
BF	$0.3005 \pm 3.160\times10^{-4}$	9.7 ± 2.05	-0.3%	0%	20
MS-BF	$0.3005 \pm 3.275\times10^{-4}$	9.8 ± 2.26	-0.3%	0%	20
SA	$0.3005 \pm 3.275\times10^{-4}$	9.8 ± 2.26	-0.3%	0%	20
Tabu	$0.3005 \pm 3.275\times10^{-4}$	9.8 ± 2.26	-0.3%	0%	20
snr+10dB_rho0.5, $\mu = 0.3$					
ALS ($R = 20$)	$0.2987 \pm 2.874\times10^{-4}$	20 ± 0	-0.9%	0%	20
TopK-PH	0.3645 ± 0.03444	7.5 ± 1.24	20.9%	30%	20
TopK-PH+refit	$0.3008 \pm 2.929\times10^{-4}$	7.5 ± 1.24	-0.2%	30%	20
TopK-IL	$0.3008 \pm 2.880\times10^{-4}$	7.5 ± 1.24	-0.2%	30%	20
Mag-IL	$0.3009 \pm 2.567\times10^{-4}$	7.05 ± 1.1	-0.2%	45%	20
Diag-x	$0.3004 \pm 2.754\times10^{-4}$	10.3 ± 2.11	-0.3%	0%	20
ES	$0.3008 \pm 2.879\times10^{-4}$	7.5 ± 1.24	-0.2%	30%	20
BF	$0.3008 \pm 2.879\times10^{-4}$	7.5 ± 1.24	-0.2%	30%	20
MS-BF	$0.3008 \pm 2.879\times10^{-4}$	7.5 ± 1.24	-0.2%	30%	20
SA	$0.3008 \pm 2.879\times10^{-4}$	7.5 ± 1.24	-0.2%	30%	20
Tabu	$0.3008 \pm 2.879\times10^{-4}$	7.5 ± 1.24	-0.2%	30%	20
snr+10dB_rho0.5, $\mu = 0.5$					
ALS ($R = 20$)	$0.2987 \pm 2.874\times10^{-4}$	20 ± 0	-0.9%	0%	20
TopK-PH	0.4005 ± 0.04978	6.55 ± 0.826	32.9%	60%	20
TopK-PH+refit	$0.3009 \pm 2.355\times10^{-4}$	6.55 ± 0.826	-0.2%	60%	20
TopK-IL	$0.3009 \pm 2.354\times10^{-4}$	6.55 ± 0.826	-0.2%	60%	20
Mag-IL	0.3096 ± 0.02661	6.05 ± 0.51	2.7%	75%	20
Diag-x	$0.3005 \pm 2.911\times10^{-4}$	9.85 ± 1.87	-0.3%	0%	20
ES	$0.3009 \pm 2.332\times10^{-4}$	6.55 ± 0.826	-0.2%	60%	20
BF	$0.3009 \pm 2.332\times10^{-4}$	6.55 ± 0.826	-0.2%	60%	20
MS-BF	$0.3009 \pm 2.332\times10^{-4}$	6.55 ± 0.826	-0.2%	60%	20
SA	$0.3009 \pm 2.332\times10^{-4}$	6.55 ± 0.826	-0.2%	60%	20
Tabu	$0.3009 \pm 2.332\times10^{-4}$	6.55 ± 0.826	-0.2%	60%	20
snr+15dB_rho0.5, $\mu = 0.1$					
ALS ($R = 20$)	$0.1736 \pm 1.349\times10^{-4}$	20 ± 0	-0.8%	0%	20
TopK-PH	0.2219 ± 0.02933	9.95 ± 1.64	26.8%	0%	20
TopK-PH+refit	$0.1745 \pm 1.178\times10^{-4}$	9.95 ± 1.64	-0.3%	0%	20
TopK-IL	$0.1745 \pm 1.201\times10^{-4}$	9.95 ± 1.64	-0.3%	0%	20
Mag-IL	$0.1745 \pm 1.162\times10^{-4}$	9.85 ± 1.69	-0.3%	0%	20
Diag-x	$0.1744 \pm 1.440\times10^{-4}$	11.8 ± 2.15	-0.4%	0%	20
ES	$0.1745 \pm 1.094\times10^{-4}$	9.95 ± 1.64	-0.3%	0%	20
BF	$0.1745 \pm 1.209\times10^{-4}$	9.8 ± 1.77	-0.3%	0%	20
MS-BF	$0.1745 \pm 1.094\times10^{-4}$	9.95 ± 1.64	-0.3%	0%	20
SA	$0.1745 \pm 1.094\times10^{-4}$	9.95 ± 1.64	-0.3%	0%	20
Tabu	$0.1745 \pm 1.094\times10^{-4}$	9.95 ± 1.64	-0.3%	0%	20
snr+15dB_rho0.5, $\mu = 0.3$					
ALS ($R = 20$)	$0.1736 \pm 1.349\times10^{-4}$	20 ± 0	-0.8%	0%	20
TopK-PH	0.2951 ± 0.0718	7.95 ± 1.36	68.6%	15%	20
TopK-PH+refit	$0.1747 \pm 1.070\times10^{-4}$	7.95 ± 1.36	-0.2%	15%	20

Method	Rel. error	$\hat{R}$	Excess over $\varepsilon^\star$	$\hat{R} = R_{\text{true}}$	n
TopK-IL	$0.1747 \pm 9.469\times10^{-5}$	7.95 ± 1.36	-0.2%	15%	20
Mag-IL	$0.1747 \pm 9.483\times10^{-5}$	7.65 ± 1.31	-0.2%	20%	20
Diag-x	$0.1745 \pm 1.639\times10^{-4}$	10.8 ± 2.06	-0.3%	0%	20
ES	$0.1747 \pm 9.499\times10^{-5}$	7.95 ± 1.36	-0.2%	15%	20
BF	$0.1747 \pm 9.499\times10^{-5}$	7.95 ± 1.36	-0.2%	15%	20
MS-BF	$0.1747 \pm 9.499\times10^{-5}$	7.95 ± 1.36	-0.2%	15%	20
SA	$0.1747 \pm 9.499\times10^{-5}$	7.95 ± 1.36	-0.2%	15%	20
Tabu	$0.1747 \pm 9.499\times10^{-5}$	7.95 ± 1.36	-0.2%	15%	20
snr+15dB_rho0.5, $\mu = 0.5$					
ALS ($R = 20$)	$0.1736 \pm 1.349\times10^{-4}$	20 ± 0	-0.8%	0%	20
TopK-PH	0.3531 ± 0.08105	6.75 ± 0.786	101.7%	45%	20
TopK-PH+refit	$0.1747 \pm 8.292\times10^{-5}$	6.75 ± 0.786	-0.2%	45%	20
TopK-IL	$0.1747 \pm 8.309\times10^{-5}$	6.75 ± 0.786	-0.2%	45%	20
Mag-IL	0.1825 ± 0.03457	6.2 ± 0.696	4.3%	80%	20
Diag-x	$0.1745 \pm 1.681\times10^{-4}$	10.4 ± 2.09	-0.3%	5%	20
ES	$0.1747 \pm 8.033\times10^{-5}$	6.75 ± 0.786	-0.2%	45%	20
BF	$0.1747 \pm 8.033\times10^{-5}$	6.75 ± 0.786	-0.2%	45%	20
MS-BF	$0.1747 \pm 8.033\times10^{-5}$	6.75 ± 0.786	-0.2%	45%	20
SA	$0.1747 \pm 8.033\times10^{-5}$	6.75 ± 0.786	-0.2%	45%	20
Tabu	$0.1747 \pm 8.033\times10^{-5}$	6.75 ± 0.786	-0.2%	45%	20
snr+20dB_rho0.5, $\mu = 0.1$					
ALS ($R = 20$)	$0.09872 \pm 6.848\times10^{-5}$	20 ± 0	-0.8%	0%	20
TopK-PH	0.1562 ± 0.04065	10.1 ± 1.93	57.0%	0%	20
TopK-PH+refit	$0.09921 \pm 6.061\times10^{-5}$	10.1 ± 1.93	-0.3%	0%	20
TopK-IL	$0.0992 \pm 7.694\times10^{-5}$	10.1 ± 1.93	-0.3%	0%	20
Mag-IL	$0.09922 \pm 6.074\times10^{-5}$	9.6 ± 1.73	-0.3%	0%	20
Diag-x	$0.09912 \pm 6.135\times10^{-5}$	12.3 ± 1.87	-0.4%	0%	20
ES	$0.09921 \pm 6.654\times10^{-5}$	10.1 ± 1.93	-0.3%	0%	20
BF	$0.09922 \pm 6.691\times10^{-5}$	9.65 ± 2.01	-0.3%	0%	20
MS-BF	$0.09921 \pm 6.654\times10^{-5}$	10.1 ± 1.93	-0.3%	0%	20
SA	$0.09921 \pm 6.654\times10^{-5}$	10.1 ± 1.93	-0.3%	0%	20
Tabu	$0.09921 \pm 6.654\times10^{-5}$	10.1 ± 1.93	-0.3%	0%	20
snr+20dB_rho0.5, $\mu = 0.3$					
ALS ($R = 20$)	$0.09872 \pm 6.848\times10^{-5}$	20 ± 0	-0.8%	0%	20
TopK-PH	0.2733 ± 0.0731	7.3 ± 1.53	174.7%	40%	20
TopK-PH+refit	$0.09929 \pm 6.000\times10^{-5}$	7.3 ± 1.53	-0.2%	40%	20
TopK-IL	$0.0993 \pm 5.610\times10^{-5}$	7.3 ± 1.53	-0.2%	40%	20
Mag-IL	$0.09931 \pm 4.817\times10^{-5}$	7 ± 1.12	-0.2%	40%	20
Diag-x	$0.09915 \pm 6.635\times10^{-5}$	11.2 ± 1.66	-0.3%	0%	20
ES	$0.0993 \pm 5.629\times10^{-5}$	7.3 ± 1.53	-0.2%	40%	20
BF	$0.0993 \pm 5.629\times10^{-5}$	7.3 ± 1.53	-0.2%	40%	20
MS-BF	$0.0993 \pm 5.629\times10^{-5}$	7.3 ± 1.53	-0.2%	40%	20
SA	$0.0993 \pm 5.629\times10^{-5}$	7.3 ± 1.53	-0.2%	40%	20
Tabu	$0.0993 \pm 5.629\times10^{-5}$	7.3 ± 1.53	-0.2%	40%	20
snr+20dB_rho0.5, $\mu = 0.5$					
ALS ($R = 20$)	$0.09872 \pm 6.848\times10^{-5}$	20 ± 0	-0.8%	0%	20
TopK-PH	0.3333 ± 0.0924	6.35 ± 0.587	235.0%	70%	20
TopK-PH+refit	$0.09933 \pm 3.039\times10^{-5}$	6.35 ± 0.587	-0.2%	70%	20
TopK-IL	$0.09933 \pm 3.302\times10^{-5}$	6.35 ± 0.587	-0.2%	70%	20
Mag-IL	0.1071 ± 0.03481	6.05 ± 0.394	7.7%	85%	20
Diag-x	$0.09918 \pm 7.122\times10^{-5}$	10.5 ± 1.96	-0.3%	0%	20
ES	$0.09933 \pm 3.065\times10^{-5}$	6.35 ± 0.587	-0.2%	70%	20
BF	$0.09933 \pm 3.161\times10^{-5}$	6.35 ± 0.671	-0.2%	75%	20
MS-BF	$0.09933 \pm 3.065\times10^{-5}$	6.35 ± 0.587	-0.2%	70%	20
SA	$0.09933 \pm 3.065\times10^{-5}$	6.35 ± 0.587	-0.2%	70%	20

Method	Rel. error	$\hat{R}$	Excess over $\varepsilon^\star$	$\hat{R} = R_{\text{true}}$	n
Tabu	$0.09933 \pm 3.065\times10^{-5}$	6.35 ± 0.587	-0.2%	70%	20
snr+5dB_rho0.5, $\mu = 0.1$					
ALS ($R = 20$)	$0.4852 \pm 6.274\times10^{-4}$	20 ± 0	-1.0%	0%	20
TopK-PH	0.4932 ± 0.004918	8.35 ± 1.46	0.7%	10%	20
TopK-PH+refit	$0.4887 \pm 5.778\times10^{-4}$	8.35 ± 1.46	-0.3%	10%	20
TopK-IL	$0.4887 \pm 5.668\times10^{-4}$	8.35 ± 1.46	-0.3%	10%	20
Mag-IL	$0.4887 \pm 5.961\times10^{-4}$	8.45 ± 1.67	-0.3%	15%	20
Diag-x	$0.4885 \pm 5.269\times10^{-4}$	9.3 ± 1.53	-0.3%	5%	20
ES	$0.4887 \pm 5.695\times10^{-4}$	8.35 ± 1.46	-0.3%	10%	20
BF	$0.4887 \pm 5.695\times10^{-4}$	8.35 ± 1.46	-0.3%	10%	20
MS-BF	$0.4887 \pm 5.695\times10^{-4}$	8.35 ± 1.46	-0.3%	10%	20
SA	$0.4887 \pm 5.695\times10^{-4}$	8.35 ± 1.46	-0.3%	10%	20
Tabu	$0.4887 \pm 5.695\times10^{-4}$	8.35 ± 1.46	-0.3%	10%	20
snr+5dB_rho0.5, $\mu = 0.3$					
ALS ($R = 20$)	$0.4852 \pm 6.274\times10^{-4}$	20 ± 0	-1.0%	0%	20
TopK-PH	0.5114 ± 0.02201	6.95 ± 0.605	4.4%	20%	20
TopK-PH+refit	$0.489 \pm 5.260\times10^{-4}$	6.95 ± 0.605	-0.2%	20%	20
TopK-IL	$0.489 \pm 5.299\times10^{-4}$	6.95 ± 0.605	-0.2%	20%	20
Mag-IL	0.4911 ± 0.008976	6.75 ± 0.786	0.2%	30%	20
Diag-x	$0.4886 \pm 5.470\times10^{-4}$	8.85 ± 1.63	-0.3%	10%	20
ES	$0.489 \pm 5.299\times10^{-4}$	6.95 ± 0.605	-0.2%	20%	20
BF	$0.4891 \pm 5.538\times10^{-4}$	6.8 ± 0.616	-0.2%	30%	20
MS-BF	$0.489 \pm 5.299\times10^{-4}$	6.95 ± 0.605	-0.2%	20%	20
SA	$0.489 \pm 5.299\times10^{-4}$	6.95 ± 0.605	-0.2%	20%	20
Tabu	$0.489 \pm 5.299\times10^{-4}$	6.95 ± 0.605	-0.2%	20%	20
snr+5dB_rho0.5, $\mu = 0.5$					
ALS ($R = 20$)	$0.4852 \pm 6.274\times10^{-4}$	20 ± 0	-1.0%	0%	20
TopK-PH	0.5334 ± 0.03967	6.15 ± 0.366	8.9%	85%	20
TopK-PH+refit	$0.4892 \pm 5.440\times10^{-4}$	6.15 ± 0.366	-0.2%	85%	20
TopK-IL	$0.4892 \pm 5.441\times10^{-4}$	6.15 ± 0.366	-0.2%	85%	20
Mag-IL	0.507 ± 0.04075	5.8 ± 0.616	3.5%	75%	20
Diag-x	$0.4886 \pm 5.605\times10^{-4}$	8.65 ± 1.5	-0.3%	10%	20
ES	$0.4892 \pm 5.441\times10^{-4}$	6.15 ± 0.366	-0.2%	85%	20
BF	0.496 ± 0.02253	5.95 ± 0.394	1.2%	85%	20
MS-BF	$0.4892 \pm 5.441\times10^{-4}$	6.15 ± 0.366	-0.2%	85%	20
SA	$0.4892 \pm 5.441\times10^{-4}$	6.15 ± 0.366	-0.2%	85%	20
Tabu	$0.4892 \pm 5.441\times10^{-4}$	6.15 ± 0.366	-0.2%	85%	20
snr-3dB_rho0.5, $\mu = 0.1$					
ALS ($R = 20$)	$0.8074 \pm 9.986\times10^{-4}$	20 ± 0	-1.0%	0%	20
TopK-PH	0.8168 ± 0.004923	6.3 ± 0.47	0.1%	70%	20
TopK-PH+refit	0.8145 ± 0.001063	6.3 ± 0.47	-0.2%	70%	20
TopK-IL	0.8145 ± 0.001063	6.3 ± 0.47	-0.2%	70%	20
Mag-IL	0.8145 ± 0.001063	6.3 ± 0.47	-0.2%	70%	20
Diag-x	0.8144 ± 0.001035	6.55 ± 0.605	-0.2%	50%	20
ES	0.8145 ± 0.001063	6.3 ± 0.47	-0.2%	70%	20
BF	0.8156 ± 0.004965	6.2 ± 0.523	-0.0%	70%	20
MS-BF	0.8145 ± 0.001063	6.3 ± 0.47	-0.2%	70%	20
SA	0.8145 ± 0.001063	6.3 ± 0.47	-0.2%	70%	20
Tabu	0.8145 ± 0.001063	6.3 ± 0.47	-0.2%	70%	20
snr-3dB_rho0.5, $\mu = 0.3$					
ALS ($R = 20$)	$0.8074 \pm 9.986\times10^{-4}$	20 ± 0	-1.0%	0%	20
TopK-PH	0.8183 ± 0.005199	6 ± 0	0.3%	100%	20
TopK-PH+refit	0.8146 ± 0.001015	6 ± 0	-0.1%	100%	20
TopK-IL	0.8146 ± 0.001015	6 ± 0	-0.1%	100%	20
Mag-IL	0.8166 ± 0.004629	5.85 ± 0.366	0.1%	85%	20

Method	Rel. error	$\hat{R}$	Excess over $\varepsilon^\star$	$\hat{R} = R_{\text{true}}$	n
Diag-x	0.8145 ± 0.001026	6.4 ± 0.503	-0.2%	60%	20
ES	0.8146 ± 0.001015	6 ± 0	-0.1%	100%	20
BF	0.8146 ± 0.001015	6 ± 0	-0.1%	100%	20
MS-BF	0.8146 ± 0.001015	6 ± 0	-0.1%	100%	20
SA	0.8146 ± 0.001015	6 ± 0	-0.1%	100%	20
Tabu	0.8146 ± 0.001015	6 ± 0	-0.1%	100%	20
snr-3dB_rho0.5, $\mu = 0.5$					
ALS ($R = 20$)	$0.8074 \pm 9.986\times10^{-4}$	20 ± 0	-1.0%	0%	20
TopK-PH	0.8218 ± 0.009799	5.85 ± 0.366	0.7%	85%	20
TopK-PH+refit	0.8168 ± 0.005409	5.85 ± 0.366	0.1%	85%	20
TopK-IL	0.8167 ± 0.005097	5.85 ± 0.366	0.1%	85%	20
Mag-IL	0.8326 ± 0.01521	4.9 ± 0.912	2.0%	30%	20
Diag-x	0.8146 ± 0.001051	6.2 ± 0.41	-0.2%	80%	20
ES	0.8167 ± 0.004995	5.85 ± 0.366	0.1%	85%	20
BF	0.8167 ± 0.004995	5.85 ± 0.366	0.1%	85%	20
MS-BF	0.8167 ± 0.004995	5.85 ± 0.366	0.1%	85%	20
SA	0.8167 ± 0.004995	5.85 ± 0.366	0.1%	85%	20
Tabu	0.8167 ± 0.004995	5.85 ± 0.366	0.1%	85%	20

S5 Penalty plateaus (manuscript Tables 14 and 15)

Table S5. Widest stable plateau for every dataset, $R_{\max}$ and selection rule. The number of grid values a plateau spans is the number of grid values in its μ range (Sec. S1); plateaus of fewer than three values lead to abstention.

Dataset	$R_{\max}$	Solver	Plateau $\hat{R}$	μ range	Stability	R_{true}	Exact at centre
amino	10	ES	–	–	–	–	–
amino	15	ES	3	$[1.5, 1.5]$	60%	–	–
amino	20	ES	–	–	–	–	–
amino	20	Mag-IL	–	–	–	–	–
amino	20	Diag-x	3	$[2, 2]$	60%	–	–
amino	30	SA	–	–	–	–	–
covid	10	ES	2	$[1, 1]$	75%	–	–
covid	15	ES	–	–	–	–	–
covid	20	ES	2	$[2, 2]$	70%	–	–
covid	20	Mag-IL	–	–	–	–	–
covid	20	Diag-x	4	$[0.65, 0.65]$	60%	–	–
covid	30	SA	–	–	–	–	–
kinetic	10	ES	2	$[0.4, 1]$	69%	–	–
kinetic	15	ES	2	$[1, 2]$	80%	–	–
kinetic	20	ES	2	$[1.5, 2]$	62%	–	–
kinetic	20	Mag-IL	4	$[0.4, 0.4]$	60%	–	–
kinetic	20	Diag-x	–	–	–	–	–
kinetic	30	SA	3	$[0.65, 0.8]$	60%	–	–
rho0.0_snr+0dB	20	ES	6	$[0.3, 1]$	96%	6	100%
rho0.0_snr+0dB	20	Mag-IL	6	$[0.3, 0.5]$	83%	6	100%
rho0.0_snr+0dB	20	Diag-x	19	$[0.005, 0.005]$	60%	6	0%
rho0.0_snr+10dB	20	ES	6	$[0.65, 2]$	83%	6	100%
rho0.0_snr+10dB	20	Mag-IL	6	$[0.65, 0.8]$	70%	6	70%
rho0.0_snr+10dB	20	Diag-x	6	$[2, 2]$	85%	6	85%
rho0.3_snr+0dB	20	ES	6	$[0.3, 1]$	92%	6	100%
rho0.3_snr+0dB	20	Mag-IL	6	$[0.3, 0.5]$	83%	6	90%
rho0.3_snr+0dB	20	Diag-x	6	$[1, 1]$	90%	6	90%
rho0.3_snr+10dB	20	ES	6	$[0.65, 2]$	84%	6	95%
rho0.3_snr+10dB	20	Mag-IL	–	–	–	6	–

Dataset	$R_{\max}$	Solver	Plateau $\hat{R}$	μ range	Stability	R_{true}	Exact at centre
rho0.3_snr+10dB	20	Diag-x	6	[2, 2]	75%	6	75%
rho0.5_snr+0dB	20	ES	6	[0.2, 1]	94%	6	100%
rho0.5_snr+0dB	20	Mag-IL	6	[0.2, 0.4]	88%	6	95%
rho0.5_snr+0dB	20	Diag-x	6	[1, 1]	85%	6	85%
rho0.5_snr+10dB	20	ES	6	[0.5, 1.5]	87%	6	100%
rho0.5_snr+10dB	20	Mag-IL	6	[0.4, 0.5]	72%	6	75%
rho0.5_snr+10dB	20	Diag-x	6	[2, 2]	65%	6	65%
rho0.7_snr+0dB	20	ES	6	[0.1, 0.65]	90%	6	100%
rho0.7_snr+0dB	20	Mag-IL	6	[0.1, 0.2]	88%	6	90%
rho0.7_snr+0dB	20	Diag-x	6	[0.65, 1.5]	79%	6	100%
rho0.7_snr+10dB	20	ES	6	[0.4, 1]	89%	6	95%
rho0.7_snr+10dB	20	Mag-IL	7	[0.2, 0.2]	60%	6	20%
rho0.7_snr+10dB	20	Diag-x	–	–	–	6	–
rho0.9_snr+0dB	20	ES	2	[0.8, 1]	68%	6	0%
rho0.9_snr+0dB	20	Mag-IL	–	–	–	6	–
rho0.9_snr+0dB	20	Diag-x	6	[0.1, 0.15]	60%	6	60%
rho0.9_snr+10dB	20	ES	6	[0.2, 0.3]	82%	6	85%
rho0.9_snr+10dB	20	Mag-IL	–	–	–	6	–
rho0.9_snr+10dB	20	Diag-x	8	[1, 1]	60%	6	0%
snr+0dB_rho0.5	10	ES	6	[0.05, 0.5]	88%	6	100%
snr+0dB_rho0.5	15	ES	6	[0.2, 0.65]	88%	6	100%
snr+0dB_rho0.5	20	ES	6	[0.2, 1]	94%	6	100%
snr+0dB_rho0.5	20	Mag-IL	6	[0.2, 0.4]	88%	6	95%
snr+0dB_rho0.5	20	Diag-x	6	[1, 1]	85%	6	85%
snr+0dB_rho0.5	30	SA	6	[0.5, 1.5]	88%	6	100%
snr+10dB_rho0.5	10	ES	6	[0.3, 0.8]	99%	6	100%
snr+10dB_rho0.5	15	ES	6	[0.3, 1]	88%	6	95%
snr+10dB_rho0.5	20	ES	6	[0.5, 1.5]	87%	6	100%
snr+10dB_rho0.5	20	Mag-IL	6	[0.4, 0.5]	72%	6	75%
snr+10dB_rho0.5	20	Diag-x	6	[2, 2]	65%	6	65%
snr+10dB_rho0.5	30	SA	6	[0.8, 2]	92%	6	100%
snr+15dB_rho0.5	10	ES	6	[0.3, 0.8]	97%	6	100%
snr+15dB_rho0.5	15	ES	6	[0.4, 1]	90%	6	95%
snr+15dB_rho0.5	20	ES	6	[0.65, 1.5]	98%	6	100%
snr+15dB_rho0.5	20	Mag-IL	6	[0.4, 0.65]	70%	6	80%
snr+15dB_rho0.5	20	Diag-x	6	[2, 2]	70%	6	70%
snr+15dB_rho0.5	30	SA	6	[1, 2]	98%	6	100%
snr+20dB_rho0.5	10	ES	6	[0.3, 1]	90%	6	100%
snr+20dB_rho0.5	15	ES	6	[0.4, 1]	92%	6	100%
snr+20dB_rho0.5	20	ES	6	[0.5, 1.5]	88%	6	95%
snr+20dB_rho0.5	20	Mag-IL	6	[0.5, 0.65]	75%	6	65%
snr+20dB_rho0.5	20	Diag-x	–	–	–	6	–
snr+20dB_rho0.5	30	SA	6	[0.65, 2]	85%	6	95%
snr+5dB_rho0.5	10	ES	6	[0.1, 0.8]	85%	6	100%
snr+5dB_rho0.5	15	ES	6	[0.3, 1]	89%	6	100%
snr+5dB_rho0.5	20	ES	6	[0.4, 1]	90%	6	100%
snr+5dB_rho0.5	20	Mag-IL	6	[0.4, 0.5]	70%	6	75%
snr+5dB_rho0.5	20	Diag-x	–	–	–	6	–
snr+5dB_rho0.5	30	SA	6	[0.65, 2]	82%	6	95%
snr-3dB_rho0.5	10	ES	6	[0.01, 0.3]	87%	6	100%
snr-3dB_rho0.5	15	ES	6	[0.05, 0.5]	86%	6	95%
snr-3dB_rho0.5	20	ES	6	[0.1, 0.65]	86%	6	100%
snr-3dB_rho0.5	20	Mag-IL	6	[0.1, 0.3]	84%	6	95%
snr-3dB_rho0.5	20	Diag-x	6	[0.2, 0.8]	76%	6	80%
snr-3dB_rho0.5	30	SA	6	[0.05, 1]	80%	6	100%

Dataset	$R_{\max}$	Solver	Plateau $\hat{R}$	μ range	Stability	R_{true}	Exact at centre
synth_default	10	ES	6	$[0.02, 0.3]$	90%	6	100%
synth_default	15	ES	6	$[0.05, 0.4]$	85%	6	100%
synth_default	20	ES	6	$[0.15, 0.65]$	86%	6	100%
synth_default	20	Mag-IL	6	$[0.15, 0.2]$	82%	6	90%
synth_default	20	Diag-x	6	$[0.2, 0.8]$	68%	6	75%
synth_default	30	SA	6	$[0.3, 0.8]$	95%	6	100%

S6 Noise-calibrated starting penalty

Table S6. $\mu_0 = \kappa\,\hat{\sigma}^2(\sum_n I_n)R_{\max}/\|\mathcal{X}\|_F^2$ and the rank exact search selects at μ_0.

Dataset	κ	$\hat{\sigma}$	σ	μ_0	$\hat{R}$	Exact
synth_default	0.5	0.00995	0.01	0.00335	20 ± 0	0%
synth_default	1	0.00995	0.01	0.00671	20 ± 0	0%
synth_default	2	0.00995	0.01	0.0134	20 ± 0	0%
synth_default	4	0.00995	0.01	0.0268	6.85 ± 0.933	45%
synth_default	8	0.00995	0.01	0.0537	6.8 ± 0.894	45%
amino	0.5	1.97	–	4.51×10^{-6}	20 ± 0	–
amino	1	1.97	–	9.03×10^{-6}	20 ± 0	–
amino	2	1.97	–	1.81×10^{-5}	20 ± 0	–
amino	4	1.97	–	3.61×10^{-5}	20 ± 0	–
amino	8	1.97	–	7.22×10^{-5}	20 ± 0	–
kinetic	0.5	19.4	–	1.81×10^{-6}	20 ± 0	–
kinetic	1	19.4	–	3.62×10^{-6}	20 ± 0	–
kinetic	2	19.4	–	7.23×10^{-6}	20 ± 0	–
kinetic	4	19.4	–	1.45×10^{-5}	20 ± 0	–
kinetic	8	19.4	–	2.89×10^{-5}	20 ± 0	–
covid	0.5	0.412	–	0.0109	20 ± 0	–
covid	1	0.412	–	0.0218	20 ± 0	–
covid	2	0.412	–	0.0436	20 ± 0	–
covid	4	0.412	–	0.0873	20 ± 0	–
covid	8	0.412	–	0.175	19.8 ± 0.444	–
snr-3dB_rho0.5	0.5	0.00975	0.00981	0.00329	20 ± 0	0%
snr-3dB_rho0.5	1	0.00975	0.00981	0.00659	20 ± 0	0%
snr-3dB_rho0.5	2	0.00975	0.00981	0.0132	20 ± 0	0%
snr-3dB_rho0.5	4	0.00975	0.00981	0.0263	6.75 ± 0.851	45%
snr-3dB_rho0.5	8	0.00975	0.00981	0.0527	6.5 ± 0.607	55%
snr+0dB_rho0.5	0.5	0.0069	0.00694	0.00247	20 ± 0	0%
snr+0dB_rho0.5	1	0.0069	0.00694	0.00495	20 ± 0	0%
snr+0dB_rho0.5	2	0.0069	0.00694	0.00989	20 ± 0	0%
snr+0dB_rho0.5	4	0.0069	0.00694	0.0198	7.2 ± 1.01	25%
snr+0dB_rho0.5	8	0.0069	0.00694	0.0396	7.05 ± 0.759	25%
snr+5dB_rho0.5	0.5	0.00388	0.0039	0.00119	20 ± 0	0%
snr+5dB_rho0.5	1	0.00388	0.0039	0.00238	20 ± 0	0%
snr+5dB_rho0.5	2	0.00388	0.0039	0.00476	20 ± 0	0%
snr+5dB_rho0.5	4	0.00388	0.0039	0.00952	9.9 ± 2.1	5%
snr+5dB_rho0.5	8	0.00388	0.0039	0.019	9.65 ± 2.01	5%
snr+10dB_rho0.5	0.5	0.00219	0.0022	4.50×10^{-4}	20 ± 0	0%
snr+10dB_rho0.5	1	0.00219	0.0022	9.01×10^{-4}	20 ± 0	0%
snr+10dB_rho0.5	2	0.00219	0.0022	0.0018	20 ± 0	0%
snr+10dB_rho0.5	4	0.00219	0.0022	0.0036	12.8 ± 2.47	0%
snr+10dB_rho0.5	8	0.00219	0.0022	0.00721	12.3 ± 2.56	0%
snr+15dB_rho0.5	0.5	0.00123	0.00123	1.52×10^{-4}	20 ± 0	0%
snr+15dB_rho0.5	1	0.00123	0.00123	3.04×10^{-4}	20 ± 0	0%

Dataset	κ	$\hat{\sigma}$	σ	μ_0	$\hat{R}$	Exact
snr+15dB_rho0.5	2	0.00123	0.00123	6.09×10^{-4}	20 ± 0	0%
snr+15dB_rho0.5	4	0.00123	0.00123	0.00122	15.2 ± 1.35	0%
snr+15dB_rho0.5	8	0.00123	0.00123	0.00243	14.8 ± 1.68	0%
snr+20dB_rho0.5	0.5	6.92×10^{-4}	6.94×10^{-4}	4.92×10^{-5}	20 ± 0	0%
snr+20dB_rho0.5	1	6.92×10^{-4}	6.94×10^{-4}	9.84×10^{-5}	20 ± 0	0%
snr+20dB_rho0.5	2	6.92×10^{-4}	6.94×10^{-4}	1.97×10^{-4}	20 ± 0	0%
snr+20dB_rho0.5	4	6.92×10^{-4}	6.94×10^{-4}	3.94×10^{-4}	15.9 ± 1.55	0%
snr+20dB_rho0.5	8	6.92×10^{-4}	6.94×10^{-4}	7.87×10^{-4}	15.6 ± 1.54	0%

S7 Core consistency

Table S7. CORCONDIA at the plateau rank and its neighbours.

Dataset	Plateau $\hat{R}$	R	CORCONDIA (median)	min
synth_default	6	5	75.1	66.5
synth_default	6	6	38.2	26.1
synth_default	6	7	< -100 (degenerate)	< -100 (degenerate)
kinetic	2	1	100	100
kinetic	2	2	< -100 (degenerate)	< -100 (degenerate)
kinetic	2	3	< -100 (degenerate)	< -100 (degenerate)
covid	2	1	100	100
covid	2	2	100	100
covid	2	3	< -100 (degenerate)	< -100 (degenerate)
snr-3dB_rho0.5	6	5	63	54.7
snr-3dB_rho0.5	6	6	25.2	1.88
snr-3dB_rho0.5	6	7	< -100 (degenerate)	< -100 (degenerate)
snr+0dB_rho0.5	6	5	80	74.9
snr+0dB_rho0.5	6	6	61.4	51.7
snr+0dB_rho0.5	6	7	< -100 (degenerate)	< -100 (degenerate)
snr+5dB_rho0.5	6	5	91.4	88.5
snr+5dB_rho0.5	6	6	87.5	85
snr+5dB_rho0.5	6	7	< -100 (degenerate)	< -100 (degenerate)
snr+10dB_rho0.5	6	5	95	92.5
snr+10dB_rho0.5	6	6	96	95.3
snr+10dB_rho0.5	6	7	< -100 (degenerate)	< -100 (degenerate)
snr+15dB_rho0.5	6	5	96.1	92.9
snr+15dB_rho0.5	6	6	98.7	98.5
snr+15dB_rho0.5	6	7	< -100 (degenerate)	< -100 (degenerate)
snr+20dB_rho0.5	6	5	96.4	94.5
snr+20dB_rho0.5	6	6	99.6	99.5
snr+20dB_rho0.5	6	7	< -100 (degenerate)	< -100 (degenerate)

S8 Solvers on identical saved QUBOs (manuscript Table 16 and Fig. 13)

Table S8. Main bank.

Solver	Gap, median (% of $f^\star$)	Excess (% of $\|\mathcal{X}\|_F^2$)	Optimum returned	P(opt) per read	Evaluations (median)	TTS_{99} (s, median)	n
BF	0	8.29 ± 16	58%	58%	49.5	2.1×10^{-4}	480
MS-BF(M=8)	0	0.21 ± 1.34	95%	72%	351	1.8×10^{-4}	480
MS-BF(M=32)	0	0 ± 0	100%	72%	$1.35e+03$	5.1×10^{-4}	480
MS-BF(M=128)	0	0 ± 0	100%	72%	$5.38e+03$	4.6×10^{-4}	480
SA	0	0 ± 0	100%	89%	9.00×10^{5}	3.9×10^{-4}	480
Tabu	0	0.0148 ± 0.293	99%	89%	20	0.025	480
Mag-IL	0	1.02 ± 3.32	81%	–	9	n/a	480
Diag-x	47.2	26.6 ± 38.6	26%	–	9	n/a	480
QAOA	19.3	9 ± 11.9	10%	$0.0\%^{\mathrm{shot}}$	59	n/a	10

Table S9. Scaling bank: share of instances solved to optimality, by number of QUBO variables.

R	BF optimum	MS-BF(M=32) optimum	SA optimum	Tabu optimum	QAOA optimum	ES time (s)
8	65%	100%	100%	100%	100%	0.025
10	60%	100%	100%	100%	40%	0.032
12	55%	100%	100%	95%	80%	0.12
14	55%	100%	100%	100%	60%	0.082
16	55%	100%	100%	100%	60%	0.0024
18	50%	100%	100%	95%	–	0.68
20	50%	95%	100%	100%	–	4.1

S9 QAOA under the uniform non-empty rule (manuscript Table 18)

Table S10.

Dataset	Method	Rel. error	$\hat{R}$	Calls	Cancelled	Empty would win	Fallback	n
noise_0.05	ES	0.9947 ± 0.001414	1.05 ± 0.224	80	0	0	0	20
noise_0.05	QAOA	0.9947 ± 0.001414	1.05 ± 0.224	80	0	4	0	20
noise_0.1	ES	$0.9985 \pm 3.504\times10^{-4}$	1 ± 0	80	0	0	0	20
noise_0.1	QAOA	$0.9985 \pm 3.504\times10^{-4}$	1 ± 0	80	0	80	60	20

S10 First-version sweeps under one protocol (manuscript Tables 20 and 21)

Table S11.

Sweep level	Method	Rel. error	$\hat{R}$	Sweeps	n
noise_0	ALS (R=20)	$2.991\times10^{-4} \pm 1.692\times10^{-4}$	20 ± 0	200	20
noise_0	TopK-PH (K=3)	0.6857 ± 0.04226	3 ± 0	200	20
noise_0	TopK-PH (K=6)	0.3185 ± 0.1042	6 ± 0	200	20
noise_0	TopK-PH (K=10)	0.08167 ± 0.06205	10 ± 0	200	20
noise_0	ES	$2.914\times10^{-5} \pm 5.800\times10^{-5}$	8.75 ± 1.68	200	20
noise_0	BF	$1.543\times10^{-5} \pm 2.837\times10^{-5}$	8.15 ± 1.9	200	20
noise_0.001	ALS (R=20)	0.1416 ± 0.00613	20 ± 0	191	20
noise_0.001	TopK-PH (K=3)	0.6834 ± 0.02789	3 ± 0	200	20
noise_0.001	TopK-PH (K=6)	0.3191 ± 0.06557	6 ± 0	200	20
noise_0.001	TopK-PH (K=10)	0.1752 ± 0.02268	10 ± 0	200	20
noise_0.001	ES	0.1424 ± 0.006186	9.85 ± 1.79	94	20
noise_0.001	BF	0.1424 ± 0.006186	9.85 ± 1.79	94	20
noise_0.01	ALS (R=20)	0.813 ± 0.01168	20 ± 0	159	20
noise_0.01	TopK-PH (K=3)	0.9024 ± 0.009615	3 ± 0	200	20
noise_0.01	TopK-PH (K=6)	0.8224 ± 0.012	6 ± 0	200	20
noise_0.01	TopK-PH (K=10)	0.8182 ± 0.01174	10 ± 0	200	20
noise_0.01	ES	0.8199 ± 0.01179	6.9 ± 0.788	79	20
noise_0.01	BF	0.8199 ± 0.01179	6.9 ± 0.788	79	20
noise_0.05	ALS (R=20)	$0.9804 \pm 9.771\times10^{-4}$	20 ± 0	121	20
noise_0.05	TopK-PH (K=3)	0.9927 ± 0.001251	3 ± 0	200	20
noise_0.05	TopK-PH (K=6)	$0.9893 \pm 8.773\times10^{-4}$	6 ± 0	200	20
noise_0.05	TopK-PH (K=10)	$0.9866 \pm 9.112\times10^{-4}$	10 ± 0	200	20
noise_0.05	ES	0.9946 ± 0.00151	1.15 ± 0.366	53	20
noise_0.05	BF	0.9946 ± 0.00151	1.15 ± 0.366	53	20
noise_0.1	ALS (R=20)	$0.9864 \pm 4.900\times10^{-4}$	20 ± 0	109	20
noise_0.1	TopK-PH (K=3)	$0.9974 \pm 6.431\times10^{-4}$	3 ± 0	200	20
noise_0.1	TopK-PH (K=6)	$0.9953 \pm 3.608\times10^{-4}$	6 ± 0	200	20
noise_0.1	TopK-PH (K=10)	$0.9926 \pm 3.971\times10^{-4}$	10 ± 0	200	20

Sweep level	Method	Rel. error	$\hat{R}$	Sweeps	n
noise_0.1	ES	$0.9985 \pm 3.504\times10^{-4}$	1 ± 0	48	20
noise_0.1	BF	$0.9985 \pm 3.504\times10^{-4}$	1 ± 0	48	20
rho_0	ALS (R=20)	$0.844 \pm 7.943\times10^{-4}$	20 ± 0	159	20
rho_0	TopK-PH (K=6)	0.8547 ± 0.006695	6 ± 0	200	20
rho_0	ES	$0.8512 \pm 9.745\times10^{-4}$	6.7 ± 0.801	65	20
rho_0	BF	$0.8512 \pm 9.745\times10^{-4}$	6.7 ± 0.801	65	20
rho_0.3	ALS (R=20)	0.8396 ± 0.003062	20 ± 0	158	20
rho_0.3	TopK-PH (K=6)	0.8498 ± 0.004664	6 ± 0	200	20
rho_0.3	ES	0.8467 ± 0.003096	7 ± 0.918	69	20
rho_0.3	BF	0.8467 ± 0.003096	7 ± 0.918	69	20
rho_0.5	ALS (R=20)	0.813 ± 0.01168	20 ± 0	159	20
rho_0.5	TopK-PH (K=6)	0.8224 ± 0.012	6 ± 0	200	20
rho_0.5	ES	0.8199 ± 0.01179	6.9 ± 0.788	79	20
rho_0.5	BF	0.8199 ± 0.01179	6.9 ± 0.788	79	20
rho_0.7	ALS (R=20)	0.7384 ± 0.02028	20 ± 0	176	20
rho_0.7	TopK-PH (K=6)	0.7472 ± 0.02097	6 ± 0	200	20
rho_0.7	ES	0.7446 ± 0.02036	7 ± 0.858	78	20
rho_0.7	BF	0.7446 ± 0.02036	7 ± 0.858	78	20
rho_0.9	ALS (R=20)	0.6166 ± 0.01236	20 ± 0	175	20
rho_0.9	TopK-PH (K=6)	0.6231 ± 0.01395	6 ± 0	200	20
rho_0.9	ES	0.6222 ± 0.0122	6.05 ± 0.826	106	20
rho_0.9	BF	0.6222 ± 0.0122	6.05 ± 0.826	105	20

S11 Split-half check of the selection protocol (manuscript Sec. 4.4)

Table S12. Selection on one half of the seeds of each distinct synthetic design, exact recovery at the selected μ on the other half.

Rule	Design	Selected on	Selected $\hat{R}$	μ	Exact on the other half
ES	snr-3dB_rho0.5	seeds 0–9	6	0.3	100%
ES	snr-3dB_rho0.5	seeds 10–19	6	0.4	100%
ES	snr+0dB_rho0.5	seeds 0–9	6	0.5	100%
ES	snr+0dB_rho0.5	seeds 10–19	6	0.5	100%
ES	snr+5dB_rho0.5	seeds 0–9	6	0.65	100%
ES	snr+5dB_rho0.5	seeds 10–19	6	0.8	100%
ES	snr+10dB_rho0.5	seeds 0–9	6	0.8	100%
ES	snr+10dB_rho0.5	seeds 10–19	6	1	100%
ES	snr+15dB_rho0.5	seeds 0–9	6	0.8	100%
ES	snr+15dB_rho0.5	seeds 10–19	6	1	100%
ES	snr+20dB_rho0.5	seeds 0–9	6	0.8	100%
ES	snr+20dB_rho0.5	seeds 10–19	6	0.8	90%
ES	synth_default	seeds 0–9	6	0.3	100%
ES	synth_default	seeds 10–19	6	0.4	100%
ES	rho0.0_snr+0dB	seeds 0–9	6	0.65	100%
ES	rho0.0_snr+0dB	seeds 10–19	6	0.65	100%
ES	rho0.0_snr+10dB	seeds 0–9	6	1	100%
ES	rho0.0_snr+10dB	seeds 10–19	6	1	100%
ES	rho0.3_snr+0dB	seeds 0–9	6	0.65	100%
ES	rho0.3_snr+0dB	seeds 10–19	6	0.5	100%
ES	rho0.3_snr+10dB	seeds 0–9	6	1	100%
ES	rho0.3_snr+10dB	seeds 10–19	6	1	90%
ES	rho0.7_snr+0dB	seeds 0–9	6	0.3	100%
ES	rho0.7_snr+0dB	seeds 10–19	6	0.3	100%
ES	rho0.7_snr+10dB	seeds 0–9	6	0.65	90%

Rule	Design	Selected on	Selected $\hat{R}$	μ	Exact on the other half
ES	rho0.7_snr+10dB	seeds 10–19	6	0.65	100%
ES	rho0.9_snr+0dB	seeds 0–9	abstain	–	–
ES	rho0.9_snr+0dB	seeds 10–19	2	0.8	0%
ES	rho0.9_snr+10dB	seeds 0–9	6	0.2	80%
ES	rho0.9_snr+10dB	seeds 10–19	abstain	–	–
Mag-IL	snr-3dB_rho0.5	seeds 0–9	6	0.15	70%
Mag-IL	snr-3dB_rho0.5	seeds 10–19	6	0.3	80%
Mag-IL	snr+0dB_rho0.5	seeds 0–9	6	0.3	90%
Mag-IL	snr+0dB_rho0.5	seeds 10–19	6	0.3	100%
Mag-IL	snr+5dB_rho0.5	seeds 0–9	abstain	–	–
Mag-IL	snr+5dB_rho0.5	seeds 10–19	abstain	–	–
Mag-IL	snr+10dB_rho0.5	seeds 0–9	6	0.5	60%
Mag-IL	snr+10dB_rho0.5	seeds 10–19	abstain	–	–
Mag-IL	snr+15dB_rho0.5	seeds 0–9	abstain	–	–
Mag-IL	snr+15dB_rho0.5	seeds 10–19	6	0.5	80%
Mag-IL	snr+20dB_rho0.5	seeds 0–9	6	0.65	70%
Mag-IL	snr+20dB_rho0.5	seeds 10–19	abstain	–	–
Mag-IL	synth_default	seeds 0–9	6	0.15	70%
Mag-IL	synth_default	seeds 10–19	abstain	–	–
Mag-IL	rho0.0_snr+0dB	seeds 0–9	abstain	–	–
Mag-IL	rho0.0_snr+0dB	seeds 10–19	6	0.4	100%
Mag-IL	rho0.0_snr+10dB	seeds 0–9	abstain	–	–
Mag-IL	rho0.0_snr+10dB	seeds 10–19	6	0.8	50%
Mag-IL	rho0.3_snr+0dB	seeds 0–9	6	0.4	90%
Mag-IL	rho0.3_snr+0dB	seeds 10–19	6	0.4	90%
Mag-IL	rho0.3_snr+10dB	seeds 0–9	abstain	–	–
Mag-IL	rho0.3_snr+10dB	seeds 10–19	abstain	–	–
Mag-IL	rho0.7_snr+0dB	seeds 0–9	6	0.15	90%
Mag-IL	rho0.7_snr+0dB	seeds 10–19	6	0.15	90%
Mag-IL	rho0.7_snr+10dB	seeds 0–9	abstain	–	–
Mag-IL	rho0.7_snr+10dB	seeds 10–19	abstain	–	–
Mag-IL	rho0.9_snr+0dB	seeds 0–9	abstain	–	–
Mag-IL	rho0.9_snr+0dB	seeds 10–19	abstain	–	–
Mag-IL	rho0.9_snr+10dB	seeds 0–9	abstain	–	–
Mag-IL	rho0.9_snr+10dB	seeds 10–19	abstain	–	–
Diag-x	snr-3dB_rho0.5	seeds 0–9	6	0.4	60%
Diag-x	snr-3dB_rho0.5	seeds 10–19	6	0.5	90%
Diag-x	snr+0dB_rho0.5	seeds 0–9	abstain	–	–
Diag-x	snr+0dB_rho0.5	seeds 10–19	abstain	–	–
Diag-x	snr+5dB_rho0.5	seeds 0–9	abstain	–	–
Diag-x	snr+5dB_rho0.5	seeds 10–19	abstain	–	–
Diag-x	snr+10dB_rho0.5	seeds 0–9	abstain	–	–
Diag-x	snr+10dB_rho0.5	seeds 10–19	abstain	–	–
Diag-x	snr+15dB_rho0.5	seeds 0–9	abstain	–	–
Diag-x	snr+15dB_rho0.5	seeds 10–19	abstain	–	–
Diag-x	snr+20dB_rho0.5	seeds 0–9	abstain	–	–
Diag-x	snr+20dB_rho0.5	seeds 10–19	13	0.01	0%
Diag-x	synth_default	seeds 0–9	6	0.4	50%
Diag-x	synth_default	seeds 10–19	6	0.65	90%
Diag-x	rho0.0_snr+0dB	seeds 0–9	abstain	–	–
Diag-x	rho0.0_snr+0dB	seeds 10–19	abstain	–	–
Diag-x	rho0.0_snr+10dB	seeds 0–9	abstain	–	–
Diag-x	rho0.0_snr+10dB	seeds 10–19	abstain	–	–
Diag-x	rho0.3_snr+0dB	seeds 0–9	abstain	–	–
Diag-x	rho0.3_snr+0dB	seeds 10–19	abstain	–	–

Rule	Design	Selected on	Selected $\hat{R}$	μ	Exact on the other half
Diag-x	rho0.3_snr+10dB	seeds 0–9	abstain	–	–
Diag-x	rho0.3_snr+10dB	seeds 10–19	abstain	–	–
Diag-x	rho0.7_snr+0dB	seeds 0–9	7	0.3	30%
Diag-x	rho0.7_snr+0dB	seeds 10–19	6	1	100%
Diag-x	rho0.7_snr+10dB	seeds 0–9	abstain	–	–
Diag-x	rho0.7_snr+10dB	seeds 10–19	abstain	–	–
Diag-x	rho0.9_snr+0dB	seeds 0–9	6	0.1	60%
Diag-x	rho0.9_snr+0dB	seeds 10–19	6	0.3	50%
Diag-x	rho0.9_snr+10dB	seeds 0–9	9	0.4	0%
Diag-x	rho0.9_snr+10dB	seeds 10–19	abstain	–	–

ES: true rank 27, wrong rank 1, abstain 2 of 30; mean 98%
Mag-IL: true rank 14, wrong rank 0, abstain 16 of 30; mean 81%
Diag-x: true rank 7, wrong rank 3, abstain 20 of 30; mean 71%

S12 Hardware and noise-model repeats (manuscript Table 19)

Table S13. One saved instance (Amino Acids, $R = 12$, $\mu = 0.1$, first pruning call), five repeats of 1024 shots per backend; angles optimized on the noiseless simulator. Noise models: fake backends of qiskit-ibm-runtime 0.50.0. Job identifiers are in `results/revision/hardware/hardware_repeats.csv`.

Backend	Calibration	Depth	Two-qubit gates	SWAPs	Optimum sampled	Gap of returned mask (%)
`ibm_fez` (device)	2026-10-29	711	528	120	1 of 5120	603 ± 155
FakeFez (noise model)	2025-02-26	710	528	120	4 of 5120	254 ± 182
FakeTorino (noise model)	2025-02-26	659	503	119	1 of 5120	60 ± 61

S13 Cost of one pruning call (manuscript Table 17)

Table S14.

Calls	Step	Mean (s)	Median (s)	n
first call ($R = R_{\max}$)	QUBO construction	0.00233	0.00231	72
first call ($R = R_{\max}$)	ES solve	0.0775	0.0781	12
first call ($R = R_{\max}$)	BF solve	0.00283	0.00279	12
first call ($R = R_{\max}$)	MS-BF solve	0.00325	0.00328	12
first call ($R = R_{\max}$)	SA solve	0.0196	0.0206	12
first call ($R = R_{\max}$)	Tabu solve	0.421	0.421	12
first call ($R = R_{\max}$)	QAOA solve	24.3	16.8	12
first call ($R = R_{\max}$)	QAOA n estimator evals	88.2	58.5	12
later calls ($R < R_{\max}$)	QUBO construction	9.42×10^{-4}	8.59×10^{-4}	216
later calls ($R < R_{\max}$)	ES solve	0.00117	0.00111	36
later calls ($R < R_{\max}$)	BF solve	0.00119	0.00116	36
later calls ($R < R_{\max}$)	MS-BF solve	0.0016	0.00161	36
later calls ($R < R_{\max}$)	SA solve	0.00866	0.00837	36
later calls ($R < R_{\max}$)	Tabu solve	0.42	0.42	36
later calls ($R < R_{\max}$)	QAOA solve	1.66	1.4	36
later calls ($R < R_{\max}$)	QAOA n estimator evals	90.5	63	36